

How many zero are in the end of 100!
We know,

$$100! = 100 \times 99 \times \dots \times 3 \times 2 \times 1$$

When two prime numbers 5×2 are multiplied, it generate zero. First of all we sub divide numbers from 1 to 100 into then sub intervals ranges from 1-10, 11-20 & so on. In each sub-interval, total no. of 5s is always less than total no. of 2s. A pair 5 & 2 only possible to generate zero. So, it is sufficient to count total 5s in each sub-interval for the desired result.

In the first range 1-10 there is total of two 5s each obtained from 5 and 10

In second range 11-20 there is total of two 5s each obtained from 15 & 20.

In third range 21-30 there is total of Three 5s, each obtained from 25 and 30

In fourth range 31-40 there is total of two 5s each obtained 35 & 40

In fifth range 41-50 there is total of three 5s, each obtained from 45 & 50,

In sixth range 51-60 there is total of two 5s each obtained from 55 & 60.

In the seventh range 61-70 there is total of two 5s, each obtained from 65 & 70.

In eighth range 71-80, there is total of three 5s, each from 75-80

In ninth range 81-90, there is total of two 5s, each from 85-90

In tenth range, 91-100 is total of three 5s & each from 95 & 100

Therefore, In each interval of 1-10, 11-20, 31-40, 41-50, 51-60, 61-70, 71-80, 81-90, 91-100 we get three 5s.

So, adding up all these numbers of 5s, we get:
 $= 2+2+2+2+2+2+3+3+3+3$
 $= 24$ zeros at end of 100.

2) 1200!

$$\text{Multiple of 5 in } 1200! = \left\lfloor \frac{1200}{5} \right\rfloor = 240$$

$$\text{Multiple of 25 in } 1200! = \left\lfloor \frac{1200}{25} \right\rfloor = 48$$

$$\text{Multiple of 125 in } 1200! = \left\lfloor \frac{1200}{125} \right\rfloor = 9$$

$$\text{Multiple of 625 in } 1200! = \left\lfloor \frac{1200}{625} \right\rfloor = 1$$

$$\therefore \text{Total zeros in end of } 1200! = 240 + 48 + 9 + 1 = 298$$

$$\begin{aligned}
 3) & 6^{78} \times 15^{150} \times 25^{53} \times 8^{27} \\
 &= (2 \times 3)^{78} \times (3 \times 5)^{150} \times (5^2)^{53} \times (2^3)^{27} \\
 &= 2^{78} \times 3^{78} \times 3^{150} \times 5^{150} \times 5^{106} \times 2^{81} \\
 &= 2^{159} \times 5^{256} \times 3^{228} \\
 &= (2 \times 5)^{159} \times 5^{99} \times 3^{228}
 \end{aligned}$$

$\therefore$ Total zeros = 157

$$4) 510! + 14^7 \times 15^{24}$$

$$\text{Multiple of 5 in } 510! = \left\lfloor \frac{510}{5} \right\rfloor = 102$$

$$\text{Multiple of 25 in } 510! = \left\lfloor \frac{510}{25} \right\rfloor = 20$$

$$\text{Multiple of 125 in } 510! = \left\lfloor \frac{510}{125} \right\rfloor = 4$$

$\therefore$ Total zeros in $510! = 126$

$$= 14^7 \times 15^{24}$$

$$= 2^7 \times 3^7 \times 5^{24} \times 3^{24}$$

$$= (2 \times 5)^7 \times 5^{17} \times 3^{31}$$

$$= 10^7 \times 5^{17} \times 3^{31}$$

$\therefore$ Total zeros in $14^7 \times 15^{24} = 7$

Q) Let, n be the positive integer, then evaluate the sum $S = 1 + 2 + 3 + \dots + n$

Solⁿ

let $f(x) = x^2$ so that $f(x+1) = (x+1)^2 = x^2 + 2x + 1$

Now,

$$f(x+1) - f(x) = (x+1)^2 - x^2 = 2x + 1$$

$$\text{for } x=1, \quad 2^2 - 1^2 = 2 \cdot 1 + 1$$

$$\text{for } x=2, \quad 3^2 - 2^2 = 2 \cdot 2 + 1$$

$$\text{for } x=3, \quad 4^2 - 3^2 = 2 \cdot 3 + 1$$

$$\text{for } x=n-1, \quad n^2 - (n-1)^2 = 2 \cdot (n-1) + 1$$

$$\text{for } x=n, \quad (n+1)^2 - n^2 = 2n + 1$$

Adding column wise,

$$(n+1)^2 - 1^2 = 2(1 + 2 + \dots + n) + n$$

$$\text{or, } n^2 + 2n = 2S + n$$

$$2S = n^2 + n$$

$$\therefore S = \frac{n(n+1)}{2}$$

Hint:-

Right hand side ko 1st Column sabai add karera,

$$2 \cdot 1 + 2 \cdot 2 + 2 \cdot 3 + \dots + 2 \cdot n$$

Right hand side ko 2nd Column sabai add karera,

$$1 + 1 + 1 + \dots + 1 = n$$

Q) let, n be positive integers, then evaluate the
Sum $(S) = 1^2 + 2^2 + 3^2 + \dots + n^2$

let, $f(x) = x^3$ so that $f(x+1) = (x+1)^3 = x^3 + 3x^2 + 3x + 1$

Now,

$$f(x+1) - f(x) = (x+1)^3 - x^3 = 3x^2 + 3x + 1$$

$$\text{for } x=1, \quad 2^3 - 1^3 = 3 \cdot 1^2 + 3 \cdot 1 + 1$$

$$\text{for } x=2, \quad 3^3 - 2^3 = 3 \cdot 2^2 + 3 \cdot 2 + 1$$

$$\text{for } x=3, \quad 4^3 - 3^3 = 3 \cdot 3^2 + 3 \cdot 3 + 1$$

$$\text{for } x=n-1, \quad n^3 - (n-1)^3 = 3(n-1)^2 + 3(n-1) + 1$$

$$\text{for } x=n, \quad (n+1)^3 - n^3 = 3n^2 + 3n + 1$$

Adding column wise,

$$(n+1)^3 - 1^3 = 3(1^2 + 2^2 + 3^2 + \dots + n^2) + 3(1 + 2 + 3 + \dots + n) + n$$

$$\text{or, } n^3 + 3n^2 + 3n + 1 - 1 = 3S + 3n(n+1) + n$$

$$\text{or, } 2(n^3 + 3n^2 + 3n) = 6S + 3n^2 + 3n + 2n$$

$$\text{or, } 2n^3 + 6n^2 + 6n = 6S + 3n^2 + 3n + 5n$$

$$\text{or, } 2n^3 + 3n^2 + n = 6S$$

$$\text{or, } n(2n^2 + 3n + 1) = 6S$$

$$\therefore S = \frac{n(n+1)(2n+1)}{6}$$

$$Q) S = 1^3 + 2^3 + 3^3 + \dots + n^3$$

$$f(x) = x^4, f(x+1) = (x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$$

$$f(x+1) - f(x) = (x+1)^4 - x^4 = 4x^3 + 6x^2 + 4x + 1$$

$$\text{for } x=1, \quad = 1^4 - 0^4 = 4 \cdot 1^3 + 6 \cdot 1^2 + 4 \cdot 1 + 1$$

$$\text{for } x=2, \quad = 2^4 - 1^4 = 4 \cdot 2^3 + 6 \cdot 2^2 + 4 \cdot 2 + 1$$

$$\text{for } x=3, \quad = 3^4 - 2^4 = 4 \cdot 3^3 + 6 \cdot 3^2 + 4 \cdot 3 + 1$$

$$\text{for } x=n, \quad (n+1)^4 - n^4 = 4n^3 + 6n^2 + 4n + 1$$

Adding Column wise;

$$(n+1)^4 - 1^4 = 4S + 6n(n+1)(2n+1) + 4n(n+1) + 1n$$

$$\text{or, } n^4 + 4n^3 + 6n^2 + 4n + 1 - 1 = 4S + (n^2 + n)(2n+1) + 2n^2 + 2n + n$$

$$\text{or, } n^4 + 4n^3 + 6n^2 + 4n = 4S + 2n^3 + n^2 + 2n^2 + n + 2n^2 + 3n$$

$$\text{or, } n^4 + 4n^3 + 6n^2 + 4n = 4S + 2n^3 + 5n^2 + 4n$$

$$\text{or, } n^4 + 4n^3 - 2n^3 + 6n^2 - 5n^2 + 4n - 4n = 4S$$

$$\text{or, } n^4 + 2n^3 + n^2 = 4S$$

$$\text{or, } n^2(n^2 + 2n + 1) = 4S$$

$$\text{or, } \frac{n^2(n+1)(n+1)}{4} = S$$

$$\therefore S = \left(\frac{n(n+1)}{2} \right)^2$$

Exercise

Q) ~~1.2.5~~ 1.2.5

Q) Suppose there are k students in the class. At the beginning of each class hour each student shakes hand with each of the other students. How many handshakes take place. If k is even, then will the no. of handshakes that take place be even or odd.

Solⁿ

→ If there is no student or only one student in the class room total no. of handshakes is 0.

→ Suppose there are two students then only 1 handshake will be possible. (1)

→ If we add another one student then total ^{students} ~~student~~ is 3. And total handshake is also 3. (1+2)

Similarly, if we add another student then total students is 4. And which results the total handshakes to be 6. (1+2+3)

A simple table illustrates the relationship between total no. of students and total no. of handshakes.

No. of students	No. of handshakes	Parity of handshakes
0	0	Even
1	0	Even
2	1	Odd
3	$1+2=3$	Odd
4	$1+2+3=6$	even
5	$1+2+3+4=10$	even

6	$1+2+3+4+5 = 15$	Odd
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7	$1+2+3+4+5+6 = 21$	Odd
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8	$1+2+3+4+5+6+7 = 28$	Even
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1		1
1		1

k	$1+2+\dots+(k-1) = \frac{(k-1)k}{2}$	
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→ From the above table, we see that first two (i.e. 0 & 1) students gives even no. of handshakes while next two rows gives odd no. of handshakes.

→ This shows that, we can't say that total no. of handshakes is even or odd according as the no. of students.

→ Conclusion:-

$d = 0, 1, 2, 3, \dots, d-1, d$

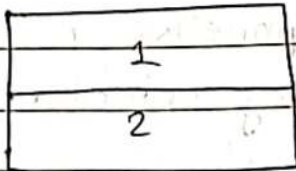
For a non-negative integers d , $4d$, $4d+1$

$4d$ & $4d+1$ students, total no. of ~~sh~~ handshakes is even.

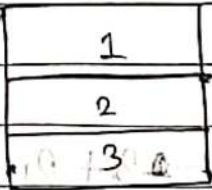
$4d+2$ & $4d+3$ students, total no. of handshakes is odd.

(1.2.6)

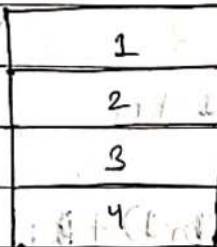
Q) What is the greatest no. of region into which three straight lines can divide the plane.



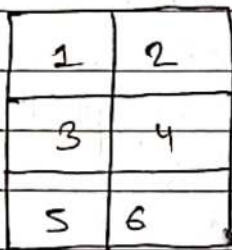
Total no. of regions = 2
(All overlapping)



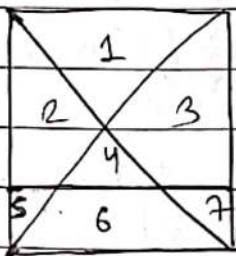
Total no. of regions = 3
(Two overlapping and one parallel)



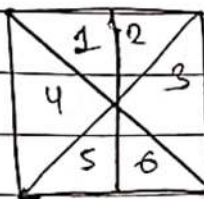
Total no. of regions = 4
(All parallel to each other)



Total no. of regions = 6
(Two parallel and other line intersecting both the lines perpendicularly)



Total no. of regions = 7
(All lines intersecting at different points)



Total no. of regions = 6
(All lines intersecting at same points)

→ Conclusion:-

→ The greatest no. of regions into which three straight lines can divide the plane is 7 when all lines are intersecting each other at different points.

* Sum of odd terms (Prove).

$$S = 1 + 3 + 5 + \dots + 2n-3 + 2n-1$$

$$\text{let } S' = 1 + 2 + 3 + \dots + n-1 + n$$

$$S + S' = 0 + 1 + 2 + \dots + n-2 + n-1 + n$$

$$2S' = 1 + 3 + 5 + \dots + (2n-3) + (2n-1) + n$$

$$\text{Q}_1, \quad \frac{2n(n+1)}{2} = S + n$$

$$\text{Q}_2, \quad n^2 + n - n = S$$

$$\boxed{S = n^2}$$

* Mathematical Induction

Q) Prove, $1+2+3+\dots+n = \frac{n(n+1)}{2}$

Let, $P(n)$:

$1+2+3+\dots+n = \frac{n(n+1)}{2}$ be proposition which is to be proved.

Step 1:-

$P(1)$ is obvious. because L.H.S. = 1 and R.H.S. = 1

Step 2:-

Assume that $P(k)$ is true i.e.

$$1+2+3+\dots+k = \frac{k(k+1)}{2}$$

Step 3:-

Now, we have to show that $P(k+1)$ is true.

i.e.

$$1+2+3+\dots+k+k+1 = \frac{(k+1)(k+1+1)}{2}$$

Now,

$$\text{L.H.S.} = \underbrace{1+2+3+\dots+k}_{P(n)} + k+1$$

$$\text{L.H.S.} = P(n) + k+1$$

$$= \frac{k(k+1)}{2} + k+1$$

$$= \frac{k^2+k+2k+2}{2}$$

$$= \frac{k(k+1)+2(k+1)}{2}$$

$$\therefore \text{L.H.S.} = \frac{(k+1)(k+2)}{2} = \text{R.H.S.}$$

2) A set with k elements has precisely 2^k subsets.

Let, $P(k)$:-

2^k be proposition which is to be proved.

Step 1:-

$P(1)$ is obvious because $2^1 = 2$, L.H.S = 2, & R.H.S = 2

A set with one element has exactly 2 subsets.

Step 2:-

Assume that $P(n)$ is true i.e.

$$P(n) = 2^n, S = \{s_1, s_2, s_3, \dots, s_n\}$$

$$S' = \{s_1, s_2, s_3, \dots, s_n\}$$

Step 3:-

Now, we have to show that $P(n+1)$ is true.

i.e.

$$P(n+1) = 2^{(n+1)} = \{s_1, s_2, s_3, \dots, s_n, s_{n+1}\}$$

$$S'' = \{s_1, s_2, s_3, \dots, s_n, s_{n+1}\}$$

Let,

$$S = \{s_1, s_2, s_3, \dots, s_n, s_{n+1}\} \quad \text{and}$$

$$S' = \{s_1, s_2, s_3, \dots, s_n\}$$

Clearly,

$S' \subset S$ and S' has 2^n subsets [from step 2]

So, the subsets of S' are subsets of S . So, initially

S has 2^n subsets. If we take union of s_{n+1}

and each of 2^n subsets of S' , again new

2^n subsets are formed, all of these are

subsets of S . Thus, we get total $2^n + 2^n$

$$2^n + 2^n = 2^{(n+1)} \text{ subsets.}$$

$\therefore P(n+1)$ is true and hence $P(k)$ is true for all $k \in \mathbb{Z}^+$.

3) Show that $7^n - 1$ is divisible by 6.

Let $P(n) \div 7^n - 1$ be ~~prop~~ divisible by 6 be proposition which is to be proved,

Step 1:-

$P(1)$ is obvious because $7^1 - 1 = 7 - 1 = 6$ which is divisible by 6.

Step 2:-

Assume that $P(k)$ is true i.e.

$P(k) = 7^k - 1$ is divisible by 6.

Step 3:-

Now, we have to show that $P(k+1)$ is true.

i.e.

$P(k+1) = 7^{(k+1)} - 1$ divisible by 6.

$$= 7^k \cdot 7 - 1$$

$$= 7(7^k - 1) + 7 - 1$$

$$= 7((7^k - 1) + 1 - 1/7)$$

$$= 7 \left(P(k) + \frac{6}{7} \right) \quad \therefore [\text{from Step 2}]$$

$$= \frac{7}{7} (7 \cdot P(k) + 6)$$

$$= 7 \cdot (P(k) + 6)$$

Here, $P(k)$ is divisible by 6 as in step 2 and obviously 6 is divisible by 6 too.

$\therefore P(k+1)$ is true and hence $P(n)$ is true for all $n \in \mathbb{Z}^+$

4) 2^n is greater than n^2 for $n \geq 5$
 $2^n > n^2$ for $n \geq 5$

Let, $P(n) : 2^n > n^2$ for $n \geq 5$ is a proposition which is to be proved.

Step 1: $P(5)$ is obvious because $2^5 = 32$ and $5^2 = 25$ and $32 > 25$ which is true.

Step 2: Assume that $P(k)$ is true i.e.
 $P(k) = 2^k > k^2$ for $k \geq 5$

Step 3: Now we have to show that $P(k+1)$ is true, i.e.

$$P(k+1) = 2^{k+1} > (k+1)^2 \text{ for } k+1 \geq 5$$

$$\text{L.H.S.} = 2^{k+1} = 2 \cdot 2^k$$

$$= 2^k + 2^k > 2^k + k^2 \quad [\text{from Step 2}]$$

$$\Rightarrow 2^k + k \cdot k > 2^k + 4k \quad [\text{Put one } k=4]$$

$$\Rightarrow 2^k + 2k + 2k > 2^k + 2k + 8 \quad [\text{Put one } k=4]$$

$$> k^2 + 2k + 8 \quad [\text{from Step 2}]$$

$$> k^2 + 2k + 1$$

$$> (k+1)^2$$

$$\therefore \text{L.H.S.} > \text{R.H.S.}$$

$$\text{i.e. } 2^{k+1} > (k+1)^2$$

$\therefore P(k+1)$ is true and hence $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Proof/Prove by Contradiction.

1) Show that $\sqrt{2}$ is irrational numbers.

$$\boxed{\begin{array}{l} p, q \neq 0, (p, q) = 1 \\ \text{of } p \text{ and } q \end{array}} \xrightarrow{\text{H.C.F.}}$$

For a no. to be rational.

Solⁿ

→ If possible suppose that $\sqrt{2}$ is a rational number then,

$$\sqrt{2} = \frac{p}{q}, \quad q \neq 0, \quad (p, q) = 1 \quad (\text{H.C.F.})$$

Squaring,

$$2 = \frac{p^2}{q^2}$$

$$\Rightarrow p^2 = 2q^2 \quad \text{--- (i)}$$

$\Rightarrow p^2$ is even.

$\Rightarrow p$ is also even.

$$\Rightarrow p = 2 \times k, \quad k \in \mathbb{Z}^+$$

from (i),

$$(2k)^2 = 2q^2$$

$$\Rightarrow q^2 = 2k^2$$

$\Rightarrow q^2$ is even

$\Rightarrow q$ is also even.

Here, both p and q are even so their G.C.D. is at least 2, which contradicts $(p, q) = 1$. So our supposition is wrong. and Hence $\sqrt{2}$ is irrational number.

2) Show that $\sqrt{7}$ is irrational number.

Solⁿ

→ If possible suppose that $\sqrt{7}$ is rational number.

$$\sqrt{7} = \frac{p}{q}, \quad q \neq 0, \quad (p, q) = 1 \quad (\text{H.C.F.})$$

Squaring,

$$7 = \frac{p^2}{q^2}$$

$$\text{or, } 7q^2 = p^2 \quad \text{--- (i)}$$

⇒ p^2 is divisible by 7

⇒ p^2 is divisible by 7^2

⇒ p is divisible by 7. (i)

$$p = 7 \times k, \quad k \in \mathbb{Z}^+$$

From (i),

$$\begin{aligned} (7k)^2 &= 7q^2 \\ 7k^2 &= q^2 \end{aligned}$$

⇒ q^2 is divisible by 7.

⇒ q is divisible by 7.

Since, p and q both can be divisible by 7, which contradicts $(p, q) = 1$. So our supposition is wrong and hence $\sqrt{7}$ is irrational number.

3) Suppose that the sum of two real numbers is 100.
Show that their products never be 3000.

Solⁿ

→ If possible suppose that x and y be two real numbers and also suppose,

$$x + y = 100 \text{ ——— (i)}$$

And

$$x \times y = 3000 \text{ ——— (ii)}$$

$$y = 100 - x$$

Now,

$$x(100 - x) = 3000$$

$$x^2 - 100x + 3000 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{100 \pm \sqrt{-2000}}{2}$$

which is complex number that contradicts x and y are real numbers. So, our supposition is wrong and hence, shows that their product never be 3000.

4) Show that there are infinitely many prime numbers.

Solⁿ

→ let us assume that there are finite number of prime numbers 'n'.

Then, the prime numbers up to n are:-

$$P_1, P_2, P_3, \dots, P_n$$

Hence, assumption of upto P_n prime numbers are taken.

Now, ~~we have~~ let us make a ~~new~~ new number by multiplying all the prime numbers and adding 1.

let, the new number be 'x'.

Now,

$$x = P_1 \times P_2 \times P_3 \times \dots \times P_n + 1$$

Here, we don't know if the number is prime or composite.

But according to our assumption there should be 'n' numbers of prime numbers. So, this new number should compulsorily be a composite number.

But if we divide our new number by the prime numbers we have, then we always get a remainder 1.

Since, it can only be divisible by 1 and the number itself and not any other prime numbers, this number should also be a prime number.

→ Hence, it contradicts our supposition as wrong and proves that there are infinitely many prime numbers.

holes

Pigeon ~~hole~~ Principle

1.4.4) Assume that k is a positive integer. If $(k+1)$ letters are delivered to k mail boxes. Then show that one mail box must contain at least two letters.

→ We prove this statement by using mathematical Induction.

Solⁿ

Let $P(k)$: If $k+1$ letters are delivered to k mail boxes, then at least one mail box must contain at least two letters.

Step 1:- $P(1)$ is (for $k=1$), $P(k)$ is obvious because there is 1 mail box and 2 letters.

Step 2:- Assume that $P(k)$ is true for $k=n$.
i.e.

If $n+1$ letters are delivered to n mail boxes then at least one mail box contains at least two letters.

Step 3:- Now, we ~~can~~ have to show that $P(k)$ is true for $k=n+1$.
i.e.

If $n+2$ letters are delivered to $n+1$ mail boxes then at least one mail box contains at least two letters.

Case I:

let last mail box is empty, then $\boxed{\text{remaining mail boxes} = n}$
 $\boxed{\text{remaining letters} = n+2}$

Then by Step 2, it is obvious that at least one box contains at least two letters or more.

Case II:

If last mail box contains exactly one letter, then, $\boxed{\text{remaining mail boxes} = n}$ and $\boxed{\text{remaining letters} = n+1}$

Then by Step 2, it clearly shows that at least one box contains at least two letters.

Case III:

If last mail box contains two or more letters, it is proved by itself.

$\therefore P(n+1)$ is true, Hence, $P(k)$ true for all k .

1.5.5) There are more adults than boys, more boys than girls, more girls than families. If no family has fewer than 3 children then what is the least no. of families that there could be?

→ Solⁿ

Let, a denotes adults, b denotes boys, g denotes girls and f represents families.

We have,

$$a > b > g > f$$

— each family has at least 3 children.

Case I:-

Let, $f = 1$ i.e. if there was only one family, there would be at least 2 girls, at least 3 boys and at least 4 adults. Four adults make at least two families, which is contradiction.

Case II:-

Let $f = 2$, i.e. if there were two families, there would be at least 3 girls, at least 4 boys and at least 5 adults. 5 adults make at least 3 families, ~~4~~ may be 2 pairs and 1 may be single, which is contradiction.

Case III :-

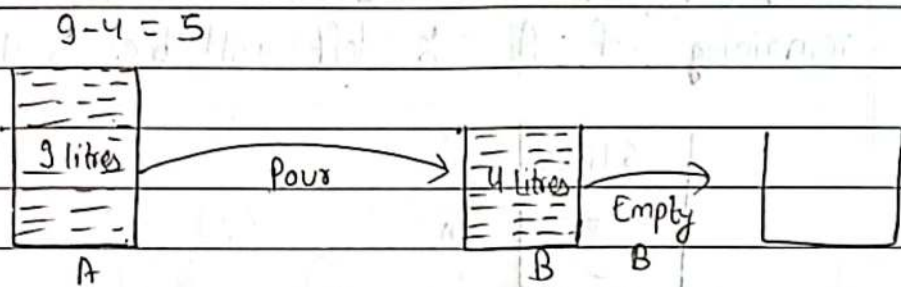
let, $f = 3$, i.e. if there were 3 families, there would be at least 4 girls, at least 5 boys and at least 6 adults. Which satisfies all given conditions. There are all total 9 children. Each family has 3 children i.e. ~~(2b, 1g)~~, (2b, 1g), (2b, 1g), (1b, 2g).

Hence, proved,,

1.6.3) Suppose that you have a 9 ^{litre} ~~gall~~ container and 4 litre container. How can you put exactly 6 litres of water into the large container?

(Exam)

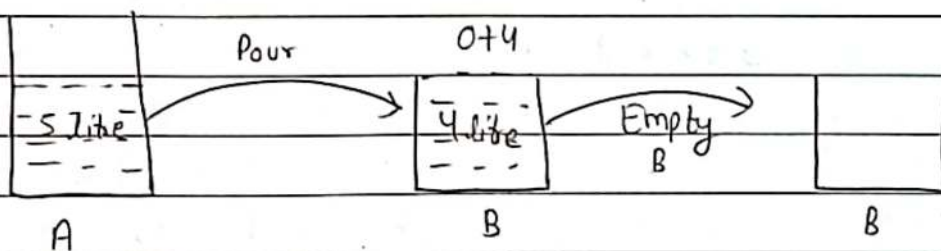
→



Fully fill

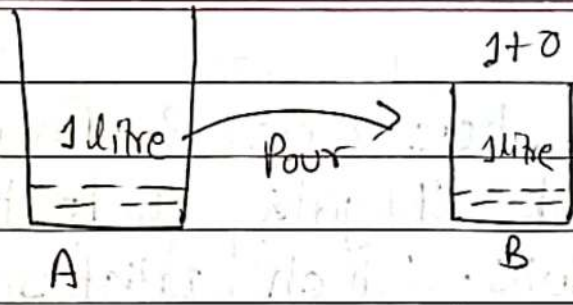
First we fully fill container A and pour B fully and empty B.

$5 - 4 = 1$



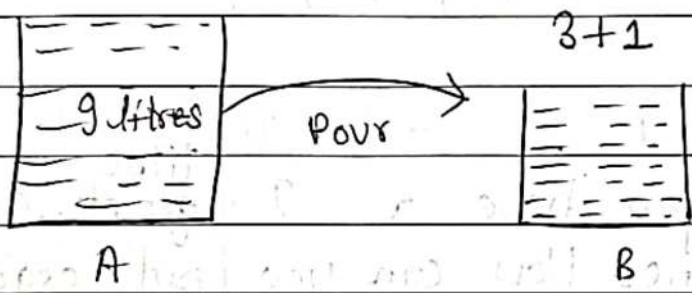
We then fill B fully from A and again empty B.

$$1 - 1 = 0$$



Now, we fill remaining of A to B and then ~~we~~ fully fill A after it is empty.

$$9 - 3 = 6$$



Now, we pour 3 litres of A into B and the remaining of A is left will be 6 litres.



1.6.5) A herd of cattle invades a barndance. Suddenly the barn is running amok with both cattle and people. A quick count reveals 120 heads and 300 feet. How many cattle are there, and how many people?

Let, x be people and y be cattle.

According to the questions,

$$x + y = 120 \quad \text{--- (i)}$$

Also, People have 2 legs and cattle have 4 legs
so,

$$2x + 4y = 300 \quad \text{--- (ii)}$$

$$x + 2y = 150 \quad \text{--- (ii)}$$

Now,

Subtracting (i) from (ii)

$$x + 2y = 150$$

$$x + y = 120$$

$$\underline{\quad - \quad - \quad}$$

$$y = 30$$

Now,

$$x = 120 - 30 = 90$$

There are 30 cattles and 90 people.

1.6.6) Suppose you are told in advance that 10 of the cattle present are lame and only have 3 feet. But the count yields 1200 and 300 feet. How ~~many~~ many cattle and how many people are there.

Solⁿ,

let, x be no. of people and y be no. of ~~no~~ cattle with 4 feet.

Since, 10 cattles have 3 feet, Total Cattles feet are 30.

Now, we subtract 30 feet from 1200. we get total = $1200 - 30 = 1170$
head'

(ii) feet total = $300 - 30 = 270$

Now, equations are:

$$x + y = 1170 \quad \text{--- (i)}$$

$$2x + 4y = 270 \quad \text{--- (ii)}$$

$$x + 2y = 135 \quad \text{--- (iii)}$$

Solving (i) and (ii),

~~$$x = 135$$~~
$$x = 85$$

$$y = 25$$

$\therefore$ There are 85 people and 25 cattles with 4 feet and 10 cattles with 3 feet total cattles 35.

1.6.9) A sheep can clear a certain field eating the grass in 1 day. A cow can clear the same field in half a day. How long does it take the sheep and the cow working together to clear the field?

Solⁿ.

Given,

A sheep can clear field eating the grass in 1 day. = 24 hrs.
A Cow can clear field eating the field in $\frac{1}{2}$ day. = 12 hrs.

In 1 day, cow can clear 2 field.

In 1 day, Cow and Sheep can clear 2 fields.

One field can be cleared in $\frac{1}{2}$ days.

$\therefore$ 8 hours one field can be cleared working together.

1.6.10) A wild beast can clear a certain field in two days. A Llama can clear in three days and a goat can clear in 4 days. How long does it take the three animals together to clear the field?

Solⁿ.

A wild beast can clear a certain field in two days.

In three days, Llama can clear a field.

In four days, goat can clear a field.

In 1 day, wild beast can clear $\frac{1}{2}$ field.

In 1 day, Llama can clear $\frac{1}{3}$ field.

In 1 day, goat can clear $\frac{1}{4}$ field.

In 1 day, all three together can complete $\frac{13}{12}$ field

1 field can be cleared by all three ~~animals~~ animals
is $\frac{12}{13}$ days.

* Unit Digit

Exercise

5) Find all pairs of Integers m, n such that $m \cdot n = m + n$

Solⁿ

We have,

$$m \cdot n = m + n$$

$$n = \frac{m + n}{m}$$

$$n = m(n-1)$$

$$\therefore m = \frac{n}{n-1}$$

Let's trial on n ,

Suppose $n=0$, then $m=0$

If $n=1$, then m is undefined.

If $n=2$, then $m=2$

Other than $n=0$ and $n=2$, if we take any integer n , then m always will be non-integer.

So, possible solutions are $(m, n) = (0, 0)$ and $(2, 2)$

14) A certain number k is a multiple of 9. Add the digits together. If the result has more than one digit, add those together. Continue adding digits together until you have one digit answer. It will be a 9. Can you explain why this is so?

Solⁿ

Soln/

Let, k is a number which is divisible by 9 having $n+1$ digits.

$$k = a_n a_{n-1} \dots a_1 a_0$$

$$\text{or, } = a_n \times 10^n + a_{n-1} \times 10^{n-1} + \dots + a_1 \times 10 + a_0$$

$$\text{or, } = a_n \times (9+1)^n + a_{n-1} \times (9+1)^{n-1} + \dots + a_1 (9+1) + a_0$$

[Recalls]

$$(a+b)^n = a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + b^n$$

divisible by 9

$$= a_n \{ 9^n + {}^nC_1 9^{n-1} + \dots + {}^nC_{n-1} 9 + 1 \} + a_{n-1} \{ 9^{n-1} + \dots + 1 \} + \dots + 9a_1 + a_1 + a_0$$

each has at least one 9

$$= a_n \{ \text{no. divisible by 9} \} + a_{n-1} \{ \text{no. divisible by 9} \} + \dots + 9a_1 + \{ a_n + a_{n-1} + \dots + a_1 + a_0 \}$$

$$= 9 \times k_1 + (a_n + a_{n-1} + \dots + a_1 + a_0)$$

Here, k and $9k_1$ both are divisible by 9. So, $(a_n + a_{n-1} + \dots + a_1 + a_0)$ must be divisible by 9.

And it is smaller than k .

Assume $(a_n + a_{n-1} + \dots + a_1 + a_0)$ be another number having more than one digit. Repeat above process until we get single digit number. The single digit number divisible by 9 has to be 9 itself.

14) A watermelon weighs 500 pounds. It turns out that 99% of the weight of the watermelon is due to water in the watermelon. After the watermelon has sat in a drying room for a while, it turns out that only 98% water by weight. How much does it weigh now?

Solⁿ

Initial weight of watermelon = 500 pounds.

% of water in it = 99%

So amount of water = 99% of 500 = 495 pounds.

Weight of melon stuffs = 500 - 495 = 5 pounds.

Let, the weight of watermelon after drying up, be x .

% water = 98%.

% of melon stuff = 2%.

Also,

Weight of melon stuff. = 2% of x .

$$\therefore, \quad 5 = 2 \times \frac{x}{100}$$

$$\therefore x = 250 \text{ pounds}$$

$$23) \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!}$$

$$= \frac{2-1}{2!} + \frac{3-1}{3!} + \frac{4-1}{4!} + \dots + \frac{(n+1)-1}{(n+1)!}$$

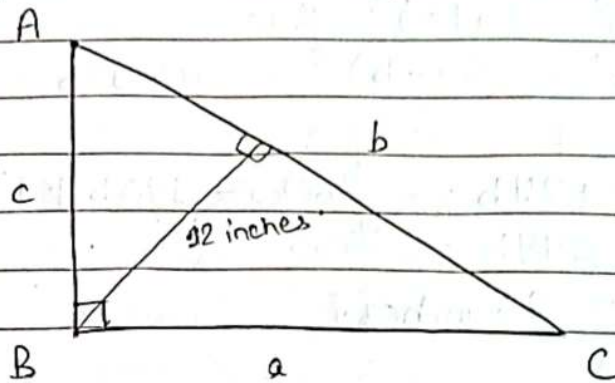
$$= 1 - \frac{1}{2!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{3!} - \dots + \frac{1}{n!} - \frac{1}{(n+1)!}$$

$$= 1 - \frac{1}{(n+1)!}$$

9)

Q 122
⑥

The perimeter of a certain right angled triangle is 60 inches. The height perpendicular to the hypotenuse is 12 inches. What are the lengths of three sides of the triangle?



Perimeter (P) = 60 inches.

Let a, b, c be sides of the right angled triangle which hypotenuse is side ' b '.

$$P = a + b + c \text{ ————— (i)}$$

By pythagores theorem,

$$b^2 = a^2 + c^2 \text{ ————— (ii)}$$

Also,

$$\text{Area of triangle} = \frac{1}{2} c \times a = \frac{1}{2} \times 12 \times b.$$

$$ca = 12b \text{ ————— (iii)}$$

Now,

From (i),

$$60 - b = a + c$$

Again, in (ii),

$$b^2 = (a+c)^2 - 2ac$$

$$\text{or } b^2 = (60-b)^2 - 2 \times 12b \quad (\text{from (iii)}),$$

$$\text{or } b^2 + 24b = 3600 - 120b + b^2$$

$$\text{or } 144b = 3600$$

$$\therefore b = 25 \text{ inches.}$$

Now, from (iii),

$$ca = 12 \times 25$$

$$\therefore c = \frac{300}{a}$$

Now, from (i),

$$60 - 25 = a + \frac{300}{a}$$

$$\text{or } 35a = a^2 + 300$$

$$\text{or } a^2 - 35a + 300 = 0.$$

• Either a

$$a = 20 \text{ inches}$$

or

$$a = 15 \text{ inches}$$

Now,

$$c = \frac{300}{20} = 15 \text{ inches}$$

$$c = \frac{300}{15} = 20 \text{ inches.}$$

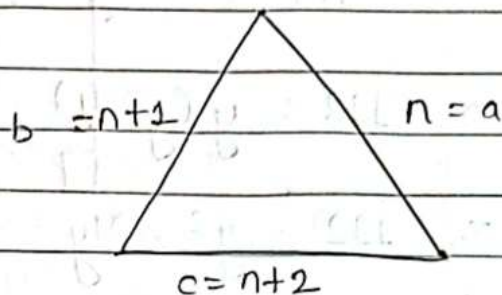
$\therefore$ The lengths of sides of triangle are 15, 20, 25 inches.

(22) The lengths of the sides of a triangle form a sequence of positive integers $\therefore$

$n, n+1, n+2$. The area of the triangle is 6.
Find the sides angle of the triangle.

Solⁿ. $a = n, b = n+1, c = n+2$.

Area of triangle (A) = 6



Now, Also,

$$A = \sqrt{s(s-a)(s-b)(s-c)} \quad \text{--- (i)}$$

$$s = \text{Semi-perimeter} = \frac{n+1+n+2}{2} = \frac{3n+3}{2}$$

Now, from (i),

$$A = \sqrt{\frac{3n+3}{2} \left(\frac{3n+3}{2} - n \right) \left(\frac{3n+3}{2} - (n+1) \right) \left(\frac{3n+3}{2} - (n+2) \right)}$$

$$\text{or, } 6^2 = \frac{3(n+1)}{2} \left[\left(\frac{n+3}{2} \right) \left(\frac{3n+3-2n-2}{2} \right) \left(\frac{3n+3-2n-4}{2} \right) \right]$$

$$\text{or, } 36 = \frac{3(n+1)}{2} \left[\left(\frac{n+3}{2} \right) \left(\frac{n+1}{2} \right) \left(\frac{n-1}{2} \right) \right]$$

Put, $n+1 = x$,

$$36 = \frac{3x}{2} \left[\left(\frac{x+2}{2} \right) \left(\frac{x}{2} \right) \left(\frac{x-2}{2} \right) \right]$$

$$\text{or, } \frac{36 \times 16}{3} = x(x-2)(x)(x-2)$$

$$\text{or, } 192 = x^2(x^2 - 4)$$

Again,

$$\text{Put, } x^2 = y.$$

$$\text{or, } 192 = y(y-4)$$

$$\text{or, } 192 = y^2 - 4y$$

$$\text{or, } y^2 - 4y - 192 = 0$$

Either,

$$y = 16$$

or,

$$y = -12$$

Since, $x^2 = y$, y can't be negative,

So,

$$y = 16.$$

Now,

$$x = \sqrt{16}$$

$$\therefore x = \pm 4 \quad \left(\begin{array}{l} \text{Since, } x \text{ is side of triangle so} \\ \therefore x = 4 \end{array} \right)$$

Since,

No negative,

$$x = n+1$$

$$\text{or, } 4 - 1 = n$$

$$\therefore n = 3$$

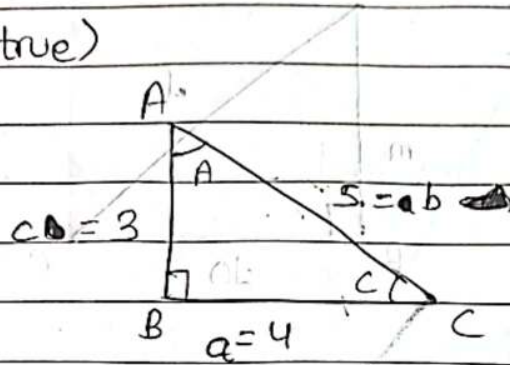
So, the Sides of triangle, are, $3, 4, 5$.
 $= 3, 4, 5$

Now, This triangle is a right angled triangle.

Since,

$$5^2 = 3^2 + 4^2$$

$$5^2 = 5^2 \text{ (true)}$$



Here,

$$\angle B = 90^\circ$$

Now, using Cosine law,

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\cos A = \frac{5^2 + 3^2 - 4^2}{2 \times 5 \times 3}$$

$$\therefore \cos A = \frac{3}{5}$$

$$\therefore A = \cos^{-1} \left(\frac{3}{5} \right) = 53.130^\circ$$

Also,

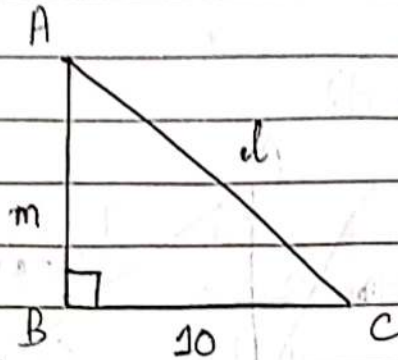
$$A + B + C = 180^\circ \text{ [Sum of triangles]}$$

$$C = 180^\circ - 53.130^\circ - 90^\circ$$

$$C = 36.87^\circ$$

ex) A right triangle has sides of length $l, m, 10$. Note that 10 is not the hypotenuse and that both l and m are integers. Find l and m .

Solⁿ.



Using pythagores, theorem,

$$l^2 = 100 + m^2 \quad \text{--- (i)}$$

Using / sine law,

Let l is hypotenuse of right angled triangle.

$$l^2 - m^2 = 100,$$

$$(l+m)(l-m) = 100,$$

Assume,

$$(l+m) = a \quad \text{--- (a)}$$

$$(l-m) = b \quad \text{--- (b)}$$

$$ab = 100 \quad \text{--- (ii)},$$

Adding (a) and (b),

$$2l = a+b$$

$$\therefore l = \frac{a+b}{2} \quad \text{--- (c)}, \quad m = \frac{a-b}{2} \quad \text{--- (d)}$$

from (ii),

$$\begin{aligned} 100 &= 100 \times 1 \\ &= \boxed{50 \times 2} \\ &= 25 \times 4 \\ &= 20 \times 5 \\ &= \boxed{10 \times 10} \end{aligned}$$

from (c) and (d) we see that, l and m are integers only when both a and b are either ~~both~~ are even or odd. We get this from, $(50, 2)$ and $(10, 10)$

If $a = b = 10$, then $m = 0$, which is not possible.

So, $a = 50$, $b = 2$ must be the numbers,

So,

$$l = \frac{50+2}{2} = 26$$

$$m = \frac{50-2}{2} = 24$$

The sides of triangle are $l = 26$, $m = 24$, 10

2.01.1) We let l and m be two lines in the plane that are skew to each other (that is, they intersect at a single point X). Let P be a point (other than X) that is on the line l . Using a ruler and compass, construct a circle that is tangent to both lines and passes through P .

Solⁿ.

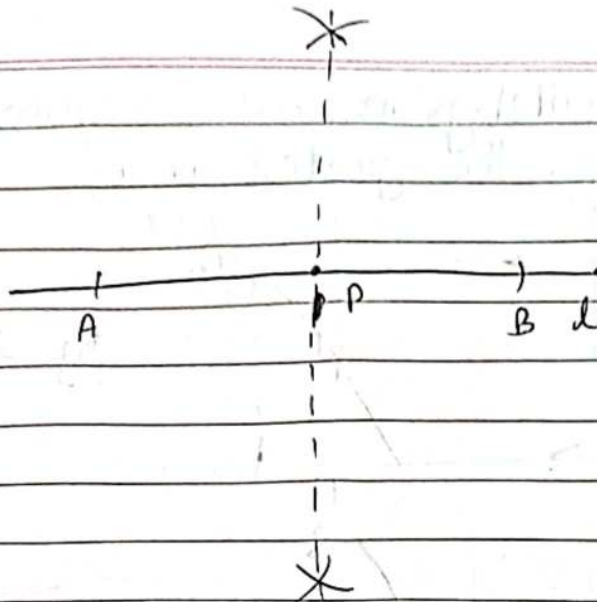
We are given that l and m ^{are} two skew lines intersected at point X . Let P be another point on line ' l '. We have to construct a circle such that whose tangent is to be at P . If we can find the centre of this circle, we can easily construct the circle.

Let Q be another point on ' m ' whose distance is same from X as from P to X .
i.e. $QX = XP$

Now, we have to construct perpendicular lines at P and Q on l and m respectively,

The intersecting point of these two perpendicular lines will be desired centre C .

* Method of constructing perpendicular line on l at P .



Locate a compass at P and strike a pencil on either side of P on A and B as shown in figure. From A and B put the non-pencil part on A and B and draw the arc above and below P. Thus creating an intersection on two points above and below. Draw the line from both the intersection on point P creating a perpendicular line at point P.

Similarly, doing the same process as on point P on line l on line m on point Q.

The perpendicular lines from point P and Q intersect at a certain point. This intersection point will be the centre of the circle having two points P and Q as their tangents.

2.3.2) Of all the parallelograms with a given perimeter, which has the greatest area?

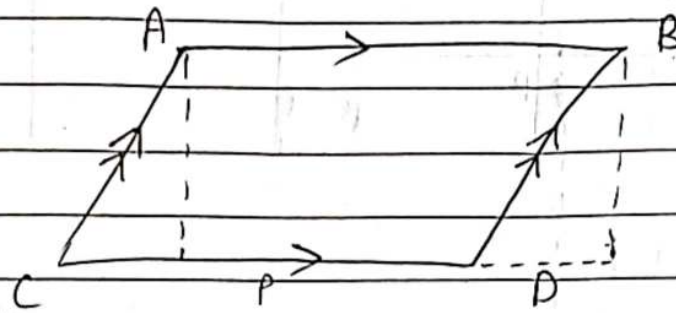


fig 2

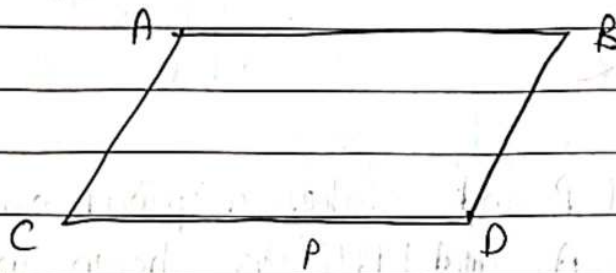


fig = 1



fig 3

let us consider a parallelogram $ABDC$ with perimeter ' P ' as shown in figure 1.

let us cut a certain part of this parallelogram and join it to another corner in order to convert slant height to vertical height as shown in figure 2.

Now, our parallelogram becomes a rectangle as shown in figure 3.

Here, perimeter will be same for all.

Suppose, the length and the breadth of the rectangle in figure is $\frac{P}{4} + \epsilon$ and $\frac{P}{4} - \epsilon$ so that its perimeter is again P .

where, ϵ is small positive quantity. $\boxed{\epsilon \geq 0}$

Then,

$$\text{Area } (A) = \left(\frac{P}{4} + \epsilon \right) \left(\frac{P}{4} - \epsilon \right)$$

$$A = \frac{P^2}{16} - \epsilon^2$$

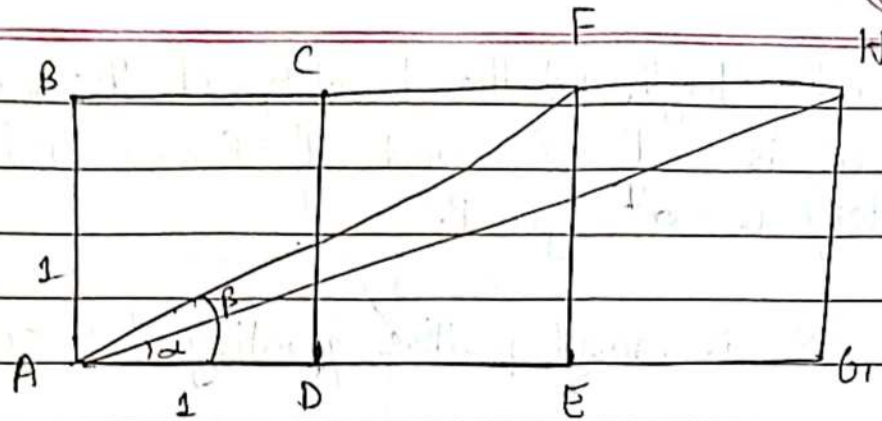
Here, A will be maximum when $\epsilon^2 = 0$.
 $\Rightarrow \epsilon = 0$

Then the length and the breadth of this rectangle be $\frac{P}{4}$, which is ultimately a square.

124)

Q

19) Look at the two angles depicted in fig 107. Show that the sum of those two angles is 45° .
[Hint: One method is use trigonometry another is to use reflection].



$$AF = \sqrt{5}$$

$$AH = \sqrt{10}$$

$$\sin \alpha = \frac{GH}{AH} = \frac{1}{\sqrt{10}} = \frac{p}{h}$$

$$\sin \beta = \frac{1}{\sqrt{5}} = \frac{p}{h}$$

Now,

$$\cos \alpha = \frac{\sqrt{10-1}}{\sqrt{10}} = \frac{3}{\sqrt{10}}$$

$$\cos \beta = \frac{\sqrt{5-1}}{\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$\begin{aligned} \sin(\alpha + \beta) &= \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta \\ &= \frac{1}{\sqrt{10}} \times \frac{2}{\sqrt{5}} + \frac{3}{\sqrt{10}} \times \frac{1}{\sqrt{5}} \end{aligned}$$

$$\alpha, \beta \quad \alpha + \beta = \sin^{-1} \left(\frac{5}{\sqrt{50}} \right) = \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$$

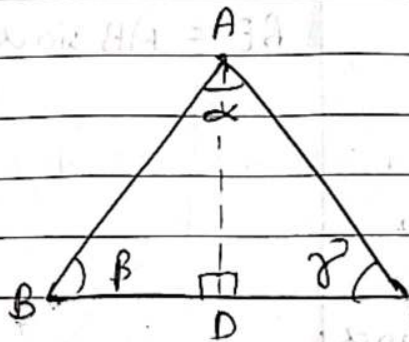
$$\therefore \alpha + \beta = 45^\circ$$

2.1.5)

2.1.10) Prove that law of sines: If $\triangle ABC$ is a triangle and α, β, γ are the angles at the vertices A, B, C respectively then,

$$\frac{\sin \alpha}{BC} = \frac{\sin \beta}{AC} = \frac{\sin \gamma}{AB}$$

Proof:



Draw $AD \perp BC$

In $\triangle ABD$,

$$\sin \beta = \frac{AD}{AB}$$

$$AD = AB \sin \beta \quad \text{--- (i)}$$

In $\triangle ADC$

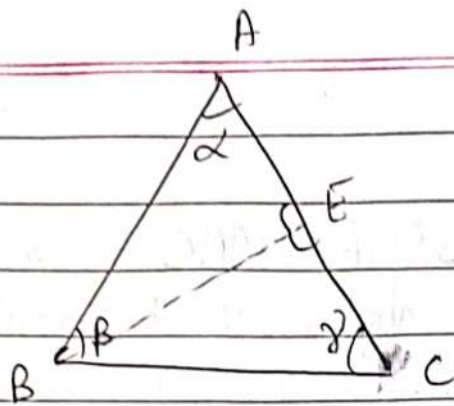
$$\sin \gamma = \frac{AD}{AC}$$

$$AD = AC \sin \gamma \quad \text{--- (ii)}$$

(i) and (ii),

$$AB \sin \beta = AC \sin \gamma$$

$$\frac{\sin \beta}{AC} = \frac{\sin \gamma}{AB} \quad \text{--- (iii)}$$



In $\triangle ABE$,

$$\sin \gamma = \frac{BE}{AB} \quad , \quad BE = AB \sin \gamma \quad \text{--- (iv)}$$

In $\triangle BEC$,

$$\sin \alpha = \frac{BE}{BC} \quad , \quad BE = BC \sin \alpha \quad \text{--- (v)}$$

from (iv) and (v),

$$AB \sin \gamma = BC \sin \alpha$$

$$\frac{\sin \gamma}{BC} = \frac{\sin \alpha}{AB} \quad \text{--- (vi)}$$

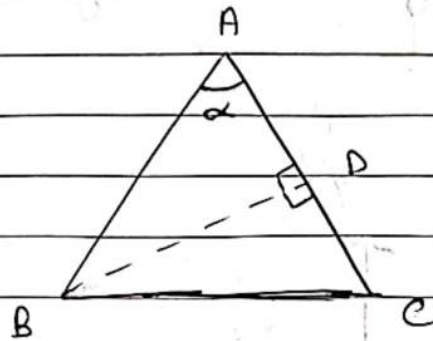
Putting (iii) & (vi) together,

$$\frac{\sin \alpha}{BC} = \frac{\sin \beta}{AC} = \frac{\sin \gamma}{AB} \quad \text{proved}$$

Q.1) Prove the law of cosine: given a triangle $\triangle ABC$, if α is the angle determined by side AB and AC , then,

$$|BC|^2 = |AB|^2 + |AC|^2 - 2|AB||AC|\cos\alpha.$$

Proof:



Draw a perpendicular line BD from B to a line AC .

In $\triangle ABD$,

$$\cos\alpha = \frac{AD}{AB}$$

$$AD = AB \cos\alpha \quad \text{--- (i)}$$

Using Pythagoras theorem, on $\triangle ABD$,

$$AB^2 = AD^2 + BD^2 \quad \text{--- (ii)}$$

$$\cancel{AB^2} = \cancel{(AC - DC)^2} + BD^2$$

Again, using Pythagoras theorem on $\triangle BDC$.

$$BC^2 = BD^2 + CD^2$$

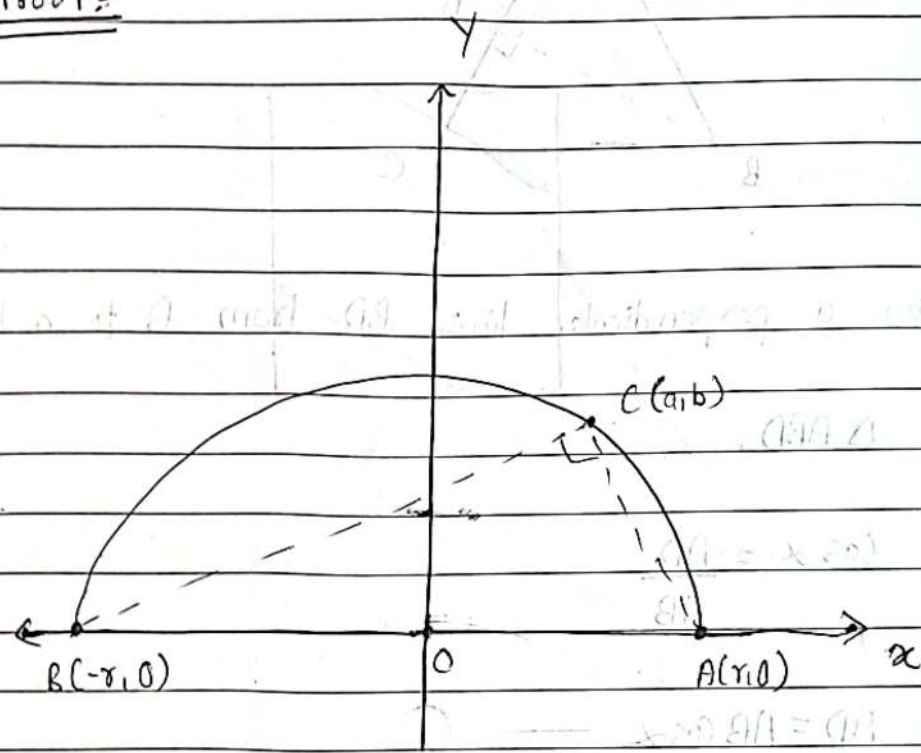
$$= BD^2 + (AC - AD)^2$$

$$BC^2 = BD^2 + AD^2 + AC^2 - 2AC \cdot AD \quad \text{--- (iii)}$$

$$BC^2 = AB^2 + AC^2 - 2 \cdot AC \cdot AB \cos \alpha \quad \text{proved,,}$$

2.2.) Prove that any angle subtended by a semi-circle is a right angle.

Proof:-



Let us consider a semi-circle, whose centre is at origin and radius r is:

$$x^2 + y^2 = r^2 \quad \text{--- (i)}$$

Let us keep this semi-circle in co-ordinate axes where $A(r, 0)$, $B(-r, 0)$ and $C(a, b)$ be three points on this circle.

Now,

$$\text{Slope of BC } (m_1) = \frac{b-0}{a-(-r)} = \frac{b}{a+r}$$

$$\text{Slope of AC } (m_2) = \frac{b}{a-r}$$

Here,

$$m_1 \cdot m_2 = \frac{b}{a+r} \times \frac{b}{a-r} \Rightarrow \frac{b^2}{a^2-r^2} \quad \text{--- (ii)}$$

Since, C(a, b) lies on a semi-circle (i), So,

$$a^2 + b^2 = r^2 \Rightarrow a^2 - r^2 = -b^2$$

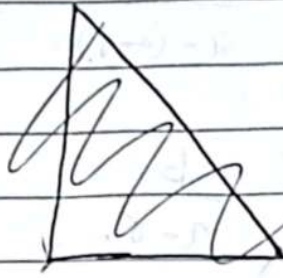
Then from (ii),

$$m_1 \cdot m_2 = \frac{b^2}{-b^2} = -1$$

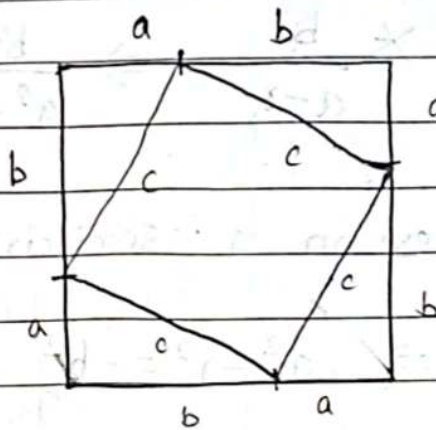
This shows that $BC \perp AC$

$\therefore$ Hence, any angle subtended by a semi-circle is a right angle.

2-1-9) Prove the pythagorean theorem.



Proof:



Let us consider a outer square whose side is of length $a+b$,
So, its area $= (a+b)^2$

Construct an inner smaller square of side c as shown in figure.

In this figure, we have four right angled triangle of equal size.

Then, we can write,

Area of outer square $= 4 \times$ area of a triangle $+ \text{area of inner square.}$

$$(a+b)^2 = 4 \times \frac{1}{2} a \times b + c^2$$

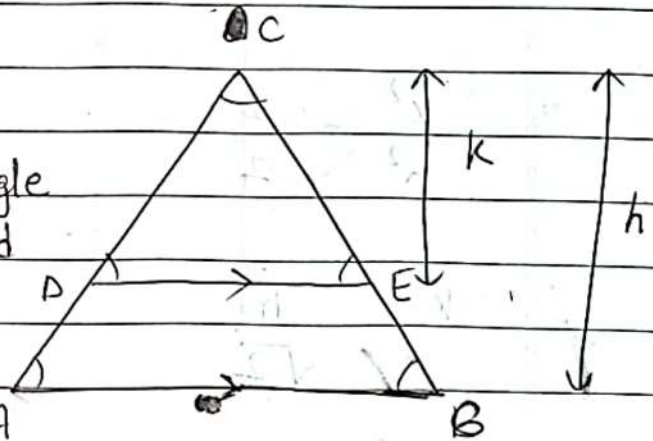
$$a^2 + b^2 + 2ab = 2ab + c^2$$

$$\therefore \boxed{a^2 + b^2 = c^2} \text{ proved,}$$

Q.5) Suppose we are given a triangle T with base $\overline{AB}$.
Give a constructive method for drawing a segment parallel to $\overline{AB}$, terminating on the sides $\overline{CA}$ and $\overline{CB}$ of the triangle that divides the triangle into two equal areas.

Solⁿ

Let $\triangle ABC$ be a given triangle T whose base is AB and height 'h'. Now we have to construct DE parallel to AB so that it divides the area of triangle into two equal halves. Let its height from C is k.



~~Similarly~~ Clearly, $\triangle CDE$ is similar to the given triangle T. So, the ratio of corresponding sides are ~~equal~~ always equal.

$$\text{i.e. } \frac{DE}{AB} = \frac{k}{h}$$

Also,

$$\frac{\text{Area of } \triangle CDE}{\text{Area of } \triangle ABC} = \frac{1}{2}$$

$$\therefore \text{Area of } \triangle CDE = \frac{1}{2} \text{ Area of } \triangle ABC$$

$$\therefore \frac{1}{2} \times DE \times k = \frac{1}{2}$$

$$\frac{1}{2} \times AB \times h$$

$$\frac{DE}{AB} = \frac{k}{h}$$

$$\therefore \frac{k}{h} \times \frac{k}{h} = \frac{1}{2}$$

$$\therefore \frac{k^2}{h^2} = \frac{1}{2}$$

$$\therefore k = \frac{h}{\sqrt{2}}$$

Therefore we have to draw DE at the height of $\frac{h}{\sqrt{2}}$ from C to divide T into two equal halves

2.16) Assume that P is a regular polygon with k sides. What is the measure of any of the k angles formed by P .

Solⁿ

Let us consider a regular polygon ' P ' with k sides and any of angles formed by ' P ' is ' α '.

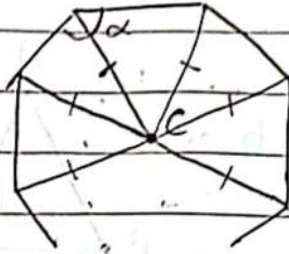


fig : P

Let ' C ' be a point inside P .

Join ' C ' with all vertices of P so that k isosceles triangles will be formed.

We know,

Sum of angles of a triangle = 180° .

$\therefore$ Total angle in P formed by k triangles = $k \cdot 180^\circ$. — (i)

On the other way,

Total angles at vertices of P = $k\alpha$

Angles formed at C = 360°

$\therefore$ Total angles in P = $k\alpha + 360^\circ$. — (ii)

Now, equating (i) and (ii),

$$k\alpha + 360 = k \cdot 180^\circ$$

$$\text{or, } k\alpha = k \cdot 180 - 360$$

$$\therefore \alpha = \frac{k \cdot 180 - 360}{k}$$

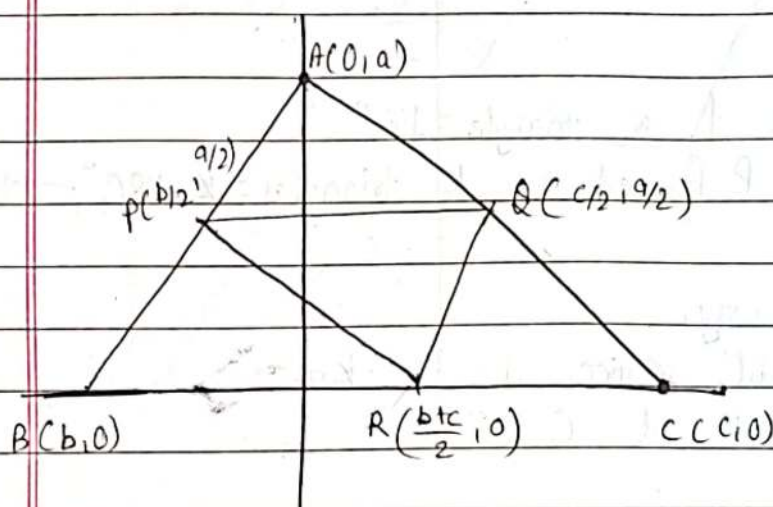
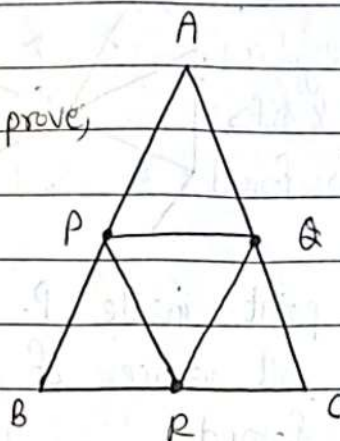
$\therefore$ The measure of any of the k angles formed by P is ' α '.

2.2.1) Suppose that 'T' is a triangle in the plane let, P, Q, R be the mid-points of each of the three sides. Prove that the triangle determined by P, Q, R is similar to the original triangle.

This is the way to prove,

Here we have to show

$$\triangle ABC \sim \triangle PQR$$



Let A, B, C be vertices of $\triangle ABC$ ~~at~~ $(a, 0), (b, 0), (c, 0)$ kept in co-ordinate axes shown in figure.

Let, $P(b/2, a/2)$, $Q(c/2, a/2)$ and $R(\frac{b+c}{2}, 0)$ are midpoints of AB, AC and BC respectively.

Here, Slope of BC = 0

$$\text{Slope of PQ} = 0 = \frac{a/2 - a/2}{b/2 - c/2} = \frac{0}{b-c} \times 2 = 0$$

$\therefore BC \parallel PQ$

Again,

$$\text{Slope of } AB = \frac{a}{-b}$$

$$\text{Slope of } QR = \frac{a/2}{-b/2} = \frac{-a}{b}$$

$AB \parallel QR$

So, $PQBR$ is a parallelogram and hence $\angle B = \angle Q$.

$$\text{Similarly, Slope of } PR = \frac{a/2 - 0}{-\frac{bc}{2}} = \frac{a}{-c}$$

$$\text{Slope of } AC = \frac{a}{-c}$$

$\therefore AC \parallel PR$

So, $AQRP$ is a parallelogram and hence $\angle A = \angle R$.

Also, $PQRC$ is a parallelogram and hence $\angle C = \angle P$.

Since,

$$\angle A = \angle R \quad [A] \quad [\text{Same angles}]$$

$$\angle B = \angle Q \quad [A] \quad [\text{Same angles}]$$

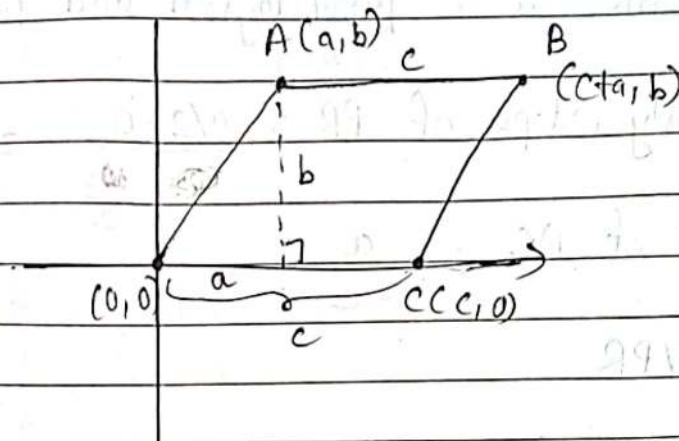
$$\angle C = \angle P \quad [A] \quad [\text{Same angles}]$$

By AAA method, we can prove that $\triangle ABC \sim \triangle PQR$.

2.24) True or False: If the diagonals of the parallelogram are perpendicular then parallelogram is a rectangle.

Sol.

Let, $OABC$ be a parallelogram whose vertices are $A(a, b)$, $O(0, 0)$, $B(cta, b)$, $C(c, 0)$ kept in a coordinate axis.



Now, Slope of diagonal $OB = \frac{b}{cta}$ (m_1)

Slope of diagonal $AC = \frac{-b}{c-a}$ (m_2)

Since, two diagonals are perpendicular according to question, So, their product must be -1 .

$$m_1 \cdot m_2 = -1$$

$$\frac{b}{cta} \times \frac{-b}{c-a} = -1$$

$$b^2 = c^2 - a^2$$

$$a^2 + b^2 = c^2$$

This shows that $OA = c$. All sides of this parallelogram are equal. But $\angle O \neq 90^\circ$.

So, it can't be a rectangle. So, false.

This is a rhombus, (Sides are equal but angles are not 90°).

2.4) Solid Geometry

$$V - E + F = 2$$

$$\text{Vertex } (V) = V$$

$$\text{Edges } (E) = \frac{3V}{2}$$

$$\text{Faces } (F) = \frac{3V}{4}$$

$\frac{3V}{4}$ (Faces meeting at each vertex)
No. of vertices to make the face.

2.4.1) Show that a polyhedron with three triangular faces meeting at each vertex must be in fact have a total of four faces.

Proof:-

Let, V, E, F be total no. of vertices, edges and faces of a polyhedron. Since three triangular faces meet at each vertex and each edge is counted twice contains two vertices, so,

$$E = \frac{3 \times V}{2}$$

$$\text{and } F = \frac{3 \times V}{3} = V. \text{ [each face contains 3 vertices]}$$

Now, using Euler's formula,

$$V - E + F = 2$$

$$V - \frac{3V}{2} + V = 2, \quad \therefore V = 4.$$

and $F = 4$ proved.)

Q.4.2) Show that if a polyhedron has all square faces, $\frac{3V}{2}$ three meeting at each vertex, then the polyhedron must be ~~square~~ ^{cube}. $\left(\frac{3V}{4}\right)$

Solⁿ.

Proof:

Let V, E, F be total no. of vertices, edges and faces of a polyhedron. Since ~~three~~ three meeting at each vertex, then must be a ~~square~~ cube.

$$E = \frac{3V}{2}$$

$$\text{and } F = \frac{3V}{4}$$

Now, using Euler's formula,

$$V - E + F = 2$$

$$V - \frac{3V}{2} + \frac{3V}{4} = 2$$

$$4V - 6V + 3V = 8$$

$$V = 8$$

$$E = \frac{3 \times 8}{2} = 12$$

$$\boxed{\text{Faces} = \frac{3 \times 8}{4} = 6}$$

Since, this polyhedron has 6 ~~faces~~ square faces, 8 vertices and 12 edges so it is a cube.

2.4.4) Explain why it is impossible to have a polyhedron with 6 triangular faces meeting at each vertex.

Solⁿ.

$$E = \frac{6V}{2} = 3V$$

$$F = \frac{6V}{3} = 2V$$

Now,

$$V - 3V + 2V = 2$$

$$0 = 2 \text{ (false)}$$

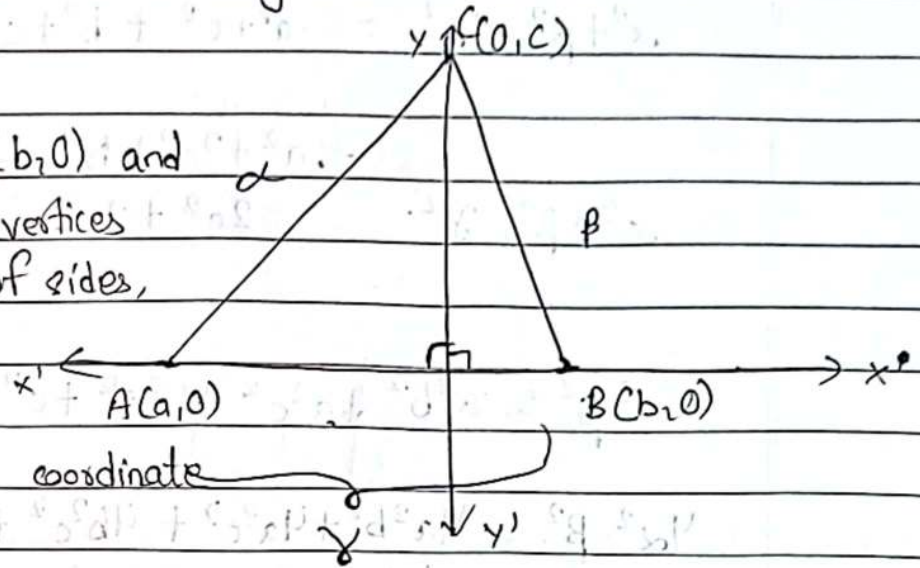
This is not possible,

So, that type of polyhedron does not exist.

1.2.3 Express the area of triangle with a formula using only the lengths of the triangle's three sides.

Solⁿ,

Let $A(a, 0)$, $B(b, 0)$ and $C(0, c)$ be vertices of triangle of sides α, β, γ .



Vertices are on coordinate axes.

$$\gamma = b + (-a) = b - a$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \gamma \times c \quad \text{--- (1)}$$

If we can express c in terms of α, β & γ , then we get the solution.

From $\triangle AOC$,

$$\alpha^2 = a^2 + c^2$$

From $\triangle BOC$,

$$\beta^2 = b^2 + c^2$$

$$\gamma = b - a$$

We have,

$$\alpha^2 + \beta^2 - \gamma^2 = a^2 + c^2 + b^2 + c^2 - \cancel{b^2} - 2a(b-c)$$

$$= a^2 + 2c^2 + b^2 - b^2 + 2ab - a^2$$

$$\alpha^2 + \beta^2 - \gamma^2 = 2c^2 + 2ab$$

$$\alpha^2 \cdot \beta^2 = a^2 b^2 + a^2 c^2 + b^2 c^2 + c^4$$

$$4\alpha^2 \cdot \beta^2 = 4a^2 b^2 + 4a^2 c^2 + 4b^2 c^2 + 4c^4 \quad \text{--- (i)}$$

$$(\alpha^2 + \beta^2 - \gamma^2)^2 = [2(c^2 + ab)]^2$$

$$= 4(c^4 + 2c^2 ab + a^2 b^2) \quad \text{--- (ii)}$$

Subtracting (ii) from (i),

$$4\alpha^2 \beta^2 - (\alpha^2 + \beta^2 - \gamma^2)^2 =$$

$$= 4a^2 c^2 + 4b^2 c^2 - 8c^2 ab$$

$$= 4c^2(a^2 - 2ab + b^2)$$

$$= 4c^2(a-b)^2$$

$$= 4c^2(-\gamma)^2$$

$$4\alpha^2 \beta^2 - \cancel{4c^2(-\gamma)^2} - (\alpha^2 + \beta^2 - \gamma^2)^2 = 4c^2(\gamma)^2$$

$$c^2 = \frac{4\alpha^2 \beta^2 - (\alpha^2 + \beta^2 - \gamma^2)^2}{4\gamma^2}$$

$$\therefore c = \sqrt{\frac{4\alpha^2\beta^2 - (\alpha^2 + \beta^2 - \gamma^2)^2}{4\gamma^2}}$$

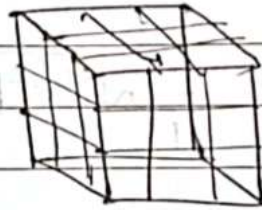
Now,

$$\text{Area of } \triangle ABC = \frac{1}{2} \gamma \times \sqrt{\frac{4\alpha^2\beta^2 - (\alpha^2 + \beta^2 - \gamma^2)^2}{4\gamma^2}}$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{4} \sqrt{4\alpha^2\beta^2 - (\alpha^2 + \beta^2 - \gamma^2)^2}$$

$$\text{Area of } \triangle ABC = \sqrt{s(s-a)(s-b)(s-c)}$$

2.4.7) Consider a unit cube with four of its eight vertices joined to form a regular tetrahedron with vertices A, B, C, D . (Figure 96). What is the ratio of the surface area of the cube to the surface area of the tetrahedron?



let us consider a cube of side 1 m
So,

$$\text{Total surface Area (A)} = 6 \times 1^2 = 6 \text{ m}^2$$

When ~~three~~ ^{four} vertices make a tetrahedron, the ~~four~~ ^{four} triangles are made. They all have equal side lengths i.e. 4 equilateral triangles are formed.

So, the side of the triangle ~~becomes~~ ^{is} the diagonal of one side of the cube.

So, length of side of triangle (a) using pythagoras

$$a^2 = 1^2 + 1^2$$

$$\therefore a = \sqrt{2}$$

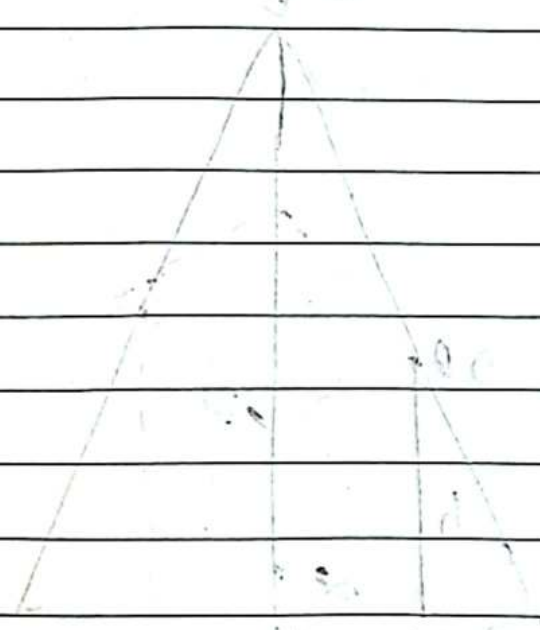
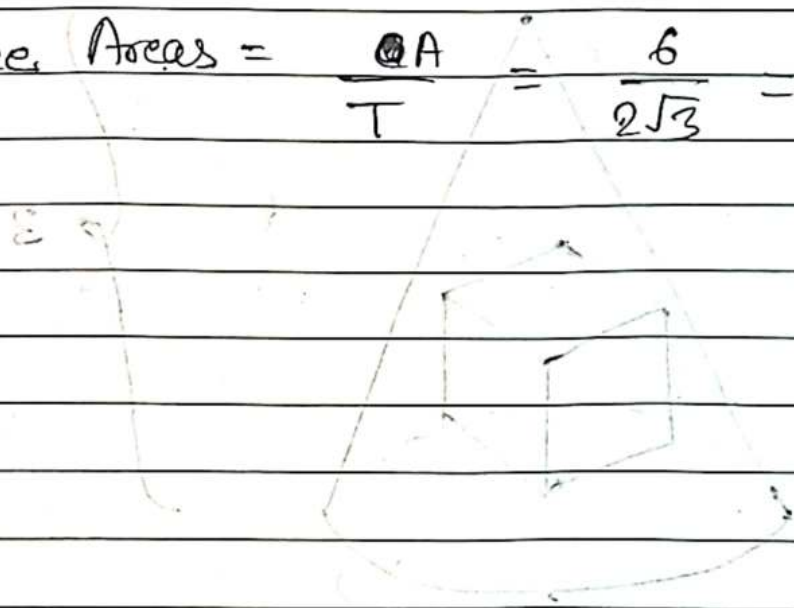
Now,

$$\text{Total surface area of Tetrahedron (T)} = \frac{4 \times \sqrt{3}}{4} \times (\sqrt{2})^2$$

$$\therefore T = 2\sqrt{3}$$

Now,

$$\text{Ratio of surface Areas} = \frac{A}{T} = \frac{6}{2\sqrt{3}} = \frac{\sqrt{3}}{1}$$



2.4.9) A right circular cone, shown in figure 98, has a cube inscribed in it. If the radius of the cone is 1, and its height is 3, then what is the volume of the cube?

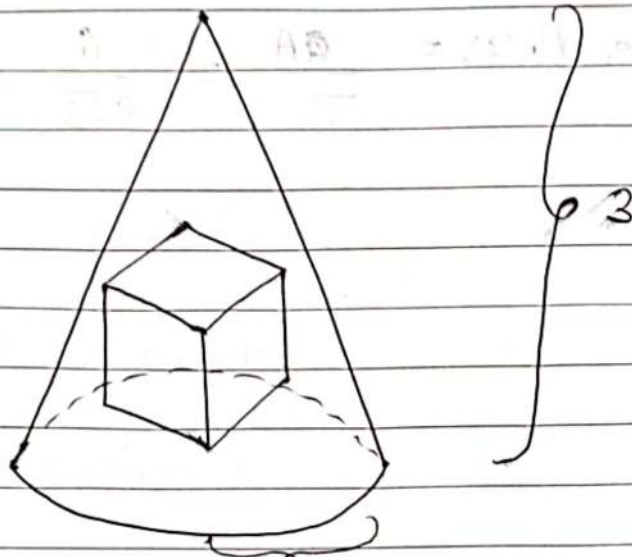


fig 98

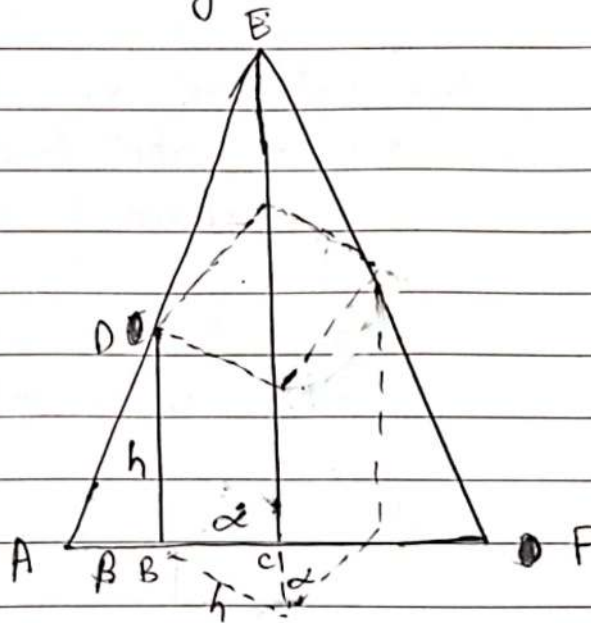


fig 99

Volume of cube $(V) = h^3$

$h = ?$

Now,

$\alpha + \beta = 1$

Suppose a cube of side 'h' is inscribed in a right circular cone of height 3 and radius 1, as shown in figure.

Let α be the half of the diagonal of the base of the cube.

Then from fig, we have,

$\alpha + \beta = 1$ ——— (i)

$\alpha^2 + \alpha^2 = h^2$

$2\alpha^2 = h^2$ ——— (ii)

From similar triangles, $\triangle ADB$ and $\triangle ACE$.

$\frac{3}{h} = \frac{1}{\beta}$

$h = 3\beta$ ——— (iii)

From (ii) & (iii),

$2\alpha^2 = (3\beta)^2$

$2\alpha^2 = 9\beta^2$

$\therefore \alpha = \frac{3\beta}{\sqrt{2}}$

Using this on (i),

$$\frac{3\beta + \beta}{\sqrt{2}} = 1.$$

$$\therefore (3 + \sqrt{2})\beta = \sqrt{2}.$$

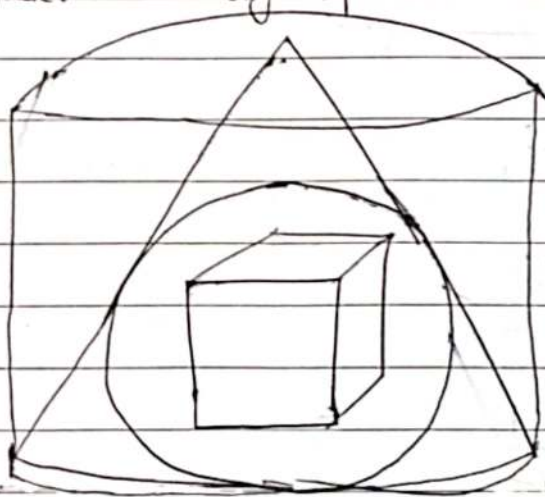
$$\therefore \frac{\sqrt{2}}{3 + \sqrt{2}} = \beta$$

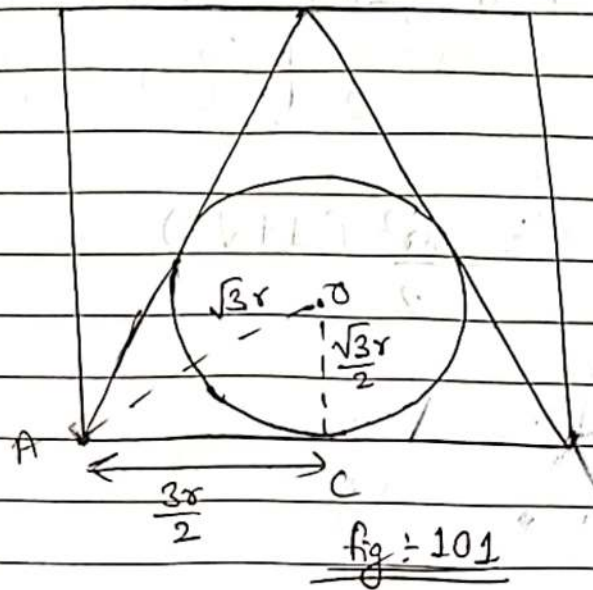
Now, from (iii),

$$(V) \text{ Volume of cube} = \left(\frac{3\sqrt{2}}{3 + \sqrt{2}} \right)^3 = \frac{54\sqrt{2}}{45\sqrt{2} + 58} \text{ cubic unit}$$

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2.4.11) A cube of side x is inscribed in a sphere. The sphere is inscribed in a cone with side length equal to the diameter of its base. The cone is inscribed in a right circular cylinder. What is the surface area of the cylinder (including top and bottom)?





Solⁿ.

Diagonal of base of cube = $\sqrt{x^2 + x^2} = \sqrt{2x^2} = \sqrt{2} \cdot x$.
Major diagonal of a cube = $\sqrt{(\sqrt{2}x)^2 + x^2} = \sqrt{3} \cdot x$.
= diameter of sphere.

From question,

The diameter of a base of cone is equal to its side length so the triangle in figure is equilateral triangle

$$AO = \sqrt{3}x$$

$$OC = \text{radius of sphere} = \frac{\sqrt{3}}{2}x$$

$$\text{Then, } AC = \sqrt{(\sqrt{3}x)^2 - \left(\frac{\sqrt{3}}{2}x\right)^2} = \sqrt{3x^2 - \frac{3x^2}{4}}$$

$$\therefore AC = \frac{3x}{2} = \text{radius of cylinder} = \text{radius}$$

Now,

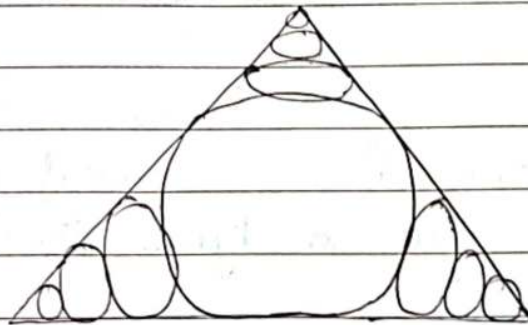
$$\text{height of cylinder} = \frac{3\sqrt{3}x}{2}$$

$$\begin{aligned} \text{Surface area of cylinder} &= 2\pi (\text{radius}) [(\text{radius}) + (\text{height})] \\ &= 2\pi \times \frac{3r}{2} \left[\frac{3r}{2} + \frac{3\sqrt{3}r}{2} \right] \end{aligned}$$

$$= \frac{9\pi r^2}{2} (1 + \sqrt{3})$$

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Q3)



$$1 + 3 \times \frac{1}{2} = \frac{5}{2}$$

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① ② ③

Total students = 41

- 12 failed in algebra
- 5 failed in biology
- 8 failed in chemistry
- 2 in algebra and biology
- 3 failed in b & c
- 1 in all subjects
- 6 in A & C

How many passed in all three subjects?

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$= 12 + 5 + 8 - 2 - 3 + 1 - 6$$

$$n(A \cup B \cup C) = 15$$

Now,

$$n(U) - n(A \cup B \cup C) = 41 - 15 = 26 \text{ failed in passed all subjects.}$$

Sure Question?

(141)

3.2-2) A ten foot pole is dropped into a milling saw and randomly cut into three shorter poles. What is the probability that these pieces will form a triangle.

Solⁿ.

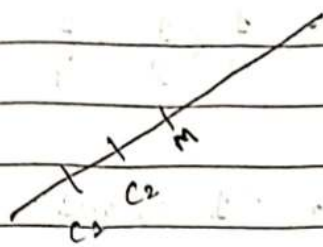
We know, the basic criteria to form a triangle is that the sum of any two sides is greater or equal to the third one. But the sum of two sides is equal to third one, then the triangle is trivial / flat triangle. This is not the case we consider.

Let, A, B, C , be the three shorter poles formed by cutting a ten foot pole. Without loss of generality $A+B > C$. Let two cuts be C_1 and C_2 and midpoint of this pole is M . If the size of any of three pieces is more than 5 feet, there is no possibility of forming triangle. Here we need two cases to form a triangle.

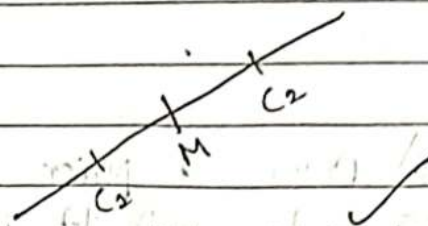
Case I:-

Let C_1 and C_2 be two cuts either both lie same side of M or ~~the~~ lie opposite side of M . Both are equally ^{likely} events, each of which has probability $\frac{1}{2}$. If both cuts lie at the same side

of M , then triangle will not be formed, so, it is necessary to have a cut at opposite sides of M . Let this event is A and $P(A) = \frac{1}{2}$.



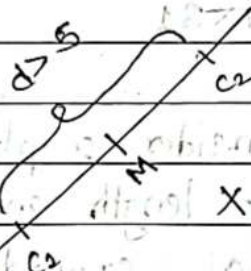
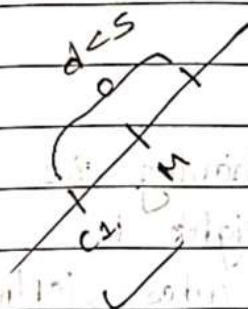
No chance of forming
triangle



Chance to form a
triangle.

Case II:-

Let the distance between two cuts be d and it is either less than or greater than 5. i.e. either $0 < d < 5$ or $5 < d < 10$. The probability of ~~occuring~~ occurrence of both event is again $\frac{1}{2}$.

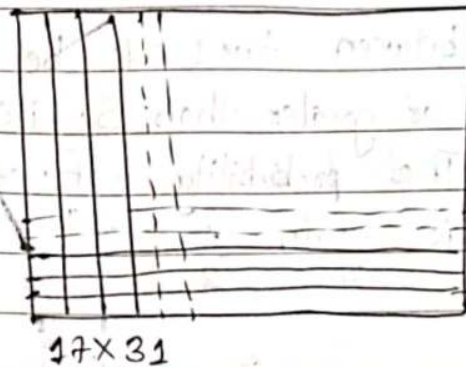


If $d > 5$, then there will be no chance to form a triangle. Let the events of having distance less than 5 is B and $P(B) = \frac{1}{2}$.

→ From above observations, we can say that both A and B are required for the formation of triangle. So, the probability that three pieces to form a triangle is:-

$$P(A \text{ and } B) = P(A) \times P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

3.5.1) Draw a planer grid that is 31 squares wide and 17 squares high. How many different non-trivial rectangles can be drawn using the lines of the grid to determine the boundaries?



let us consider a planer grid having 31 squares at length and 17 at height is kept in a co-ordinate axis whose leftmost vertex is at origin. By taking this vertex as a corner of rectangles, we get total of 31×17 rectangles. i.e. 527 rectangles.

We can calculate all rectangles taking each vertex as a leftmost vertex of rectangles by similar manner. But this is very tedious process.

let (i, j) be a random vertex where $0 < i \leq 30$ and $0 \leq j \leq 16$.

Then total non-trivial rectangles at $(i, j) = (31-i)(17-j)$

$$\begin{aligned}
 &= (31-i)(17-j) \\
 &= 527 - 17i - 31j + ij \\
 \text{Total non-trivial rectangles} &= \sum_{i=0}^{30} \sum_{j=0}^{16} (527 - 17i - 31j + ij)
 \end{aligned}$$

$$\begin{aligned}
 &= 527 \left(\sum_{i=0}^{30} 1 \right) \left(\sum_{j=0}^{16} 1 \right) - 17 \left(\sum_{i=0}^{30} i \right) \times \left(\sum_{j=0}^{16} 1 \right) \\
 &\quad - 31 \left(\sum_{i=0}^{30} 1 \right) \left(\sum_{j=0}^{16} j \right) + \left(\sum_{i=0}^{30} i \right) \left(\sum_{j=0}^{16} j \right)
 \end{aligned}$$

$$= 527 \times 31 \times 17 - 17 \times \frac{30 \times 31}{2} \times 17 - 31 \times 31 \times \frac{16 \times 17}{2} + \frac{30 \times 31}{2} \times \frac{16 \times 17}{2}$$

$$= 75888$$

33.2 Let x be a real number and k is positive integer. Calculate.

$$S = 1 + x + x^2 + \dots + x^k$$

Solⁿ.

Given,

$$S = 1 + x + x^2 + \dots + x^k \quad \text{--- (i)}$$

Multiplying both sides by x .

$$= (31-i)(17-j) \\ = 527 - 17i - 31j + ij$$

$$\text{Total non-trivial rectangles} = \sum_{i=0}^{30} \sum_{j=0}^{16} (527 - 17i - 31j + ij)$$

$$= 527 \left(\sum_{i=0}^{30} 1 \right) \left(\sum_{j=0}^{16} 1 \right) - 17 \left(\sum_{i=0}^{30} i \right) \times$$

$$\left(\sum_{j=0}^{16} 1 \right) - 31 \left(\sum_{i=0}^{30} 1 \right) \left(\sum_{j=0}^{16} j \right) + \left(\sum_{i=0}^{30} i \right) \left(\sum_{j=0}^{16} j \right)$$

$$= 527 \times 31 \times 17 - 17 \times \frac{30 \times 31}{2} \times 17 -$$

$$- 31 \times 31 \times \frac{16 \times 17}{2} + \frac{30 \times 31}{2} \times \frac{16 \times 17}{2}$$

$$= 75888$$

33.2 Let λ be a real number and k is positive integer. Calculate.

$$S = 1 + \lambda + \lambda^2 + \dots + \lambda^k$$

Solⁿ.

Given,

$$S = 1 + \lambda + \lambda^2 + \dots + \lambda^k \quad \text{--- (i)}$$

Multiplying both sides by λ .

$$S = 1 + \lambda + \lambda^2 + \lambda^3 + \dots + \lambda^{k+1} \quad \text{--- (ii)}$$

Subtracting (i) from (ii),

$$S - S = \lambda^{k+1} - 1$$

$$(\lambda - 1)S = \lambda^{k+1} - 1$$

$$\therefore S = \frac{\lambda^{k+1} - 1}{\lambda - 1}$$

When $k \rightarrow \infty$ and $|\lambda| < 1$ then

$$\text{Then } \lambda^{k+1} \rightarrow 0$$

$$S = \frac{0 - 1}{\lambda - 1}$$

$$S = \frac{1}{1 - \lambda}$$

Fibonacci Sequence.

0, 1, 1, 2, 3, 5, 8,

$$a_n = a_{n-1} + a_{n-2}$$

Show that the following formula for the fibonacci sequence is valid.

$$a_j = \frac{\left(\frac{1 + \sqrt{5}}{2}\right)^j - \left(\frac{1 - \sqrt{5}}{2}\right)^j}{\sqrt{5}}$$

Solⁿ

We know the terms of fibonacci sequence,

$a_0 = 0, a_1 = 1, a_2 = 1, a_3 = 2, a_4 = 3$ and

$$a_n = a_{n-1} + a_{n-2} \quad \text{--- (1)}$$

The characteristics eqⁿ of (i) is:-

$$r^{n-2} (r^2 - r - 1) = 0$$

$$r^2 - r - 1 = 0$$

$$r = \frac{-(-1) \pm \sqrt{1^2 - 4 \cdot 1(-1)}}{2 \times 1} = \frac{1 \pm \sqrt{5}}{2}$$

$a_n = r^n$
$a_{n-1} = r^{n-1}$
$a_{n-2} = r^{n-2}$

We get two distinct and real roots,

$$\text{let, } r_1 = \frac{1+\sqrt{5}}{2}, \quad r_2 = \frac{1-\sqrt{5}}{2}$$

$$a_n = C_1 r_1^n + C_2 r_2^n$$

$$a_n = C_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + C_2 \left(\frac{1-\sqrt{5}}{2} \right)^n \quad \text{--- (ii)}$$

Put, $n=0$,

$$a_0 = C_1 + C_2$$

$$\Rightarrow C_1 + C_2 = 0 \quad \text{--- (a)}$$

Put $n=1$,

$$a_1 = C_1 \left(\frac{1+\sqrt{5}}{2} \right) + C_2 \left(\frac{1-\sqrt{5}}{2} \right)$$

$$C_1 \left(\frac{1+\sqrt{5}}{2} \right) + C_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1 \quad \text{--- (b)}$$

Solving (a) and (b),

$$C_1 = 0.447, \quad C_2 = -0.447$$

Then from --- (ii),

$$a_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

$$\therefore a_n = \frac{\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n}{\sqrt{5}}$$

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Problem - 30

$$1 = 0 + 1$$

$$2+3+4 = 1 + 8$$

$$5+6+7+8+9 = 8 + 27$$

$$10+11+12+\dots+16 = 27 + 64$$

Patterns.

$$[(n-1)^2+1] + [(n-1)^2+2] + [(n-1)^2+3] + \dots + n^2 = (n-1)^3 + n^3$$

$$= [(n-1)^2+1] + [(n-1)^2+2] + [(n-1)^2+3] + \dots + n^2$$

$$= [(n-1)^2+1] + [(n-1)^2+2] + \dots + [(n-1)^2+(2n-1)]$$

$$= (2n-1)(n-1)^2 + [1+2+\dots+(2n-1)]$$

$$= (2n-1)(n-1)^2 + \frac{(2n-1)(2n-1+1)}{2}$$

$$= (2n-1) [(n-1)^2 + n]$$

$$= (2n-1)(n^2-n+1) = 2n^3 - 3n^2 + 3n - 1$$

$$R.H.S. = (n-1)^3 + n^3 = n^3 - 3n^2 + 3n - 1 + n^3$$

$$= 2n^3 - 3n^2 + 3n - 1$$

39)

$$1 = 1$$

$$1-4 = -(1+2)$$

$$1-4+9 = 1+2+3$$

$$1-4+9-16 = -(1+2+3+4)$$

$$1-4+9-16+\dots+(-1)^{n+1}n^2 = (-1)^{n+1}(1+2+\dots+n) \text{ pattern}$$

Proof:Case I: when n is even, then $n+1$ is odd.L.H.S.

$$1-4+9-16+\dots+(-1)^{n+1}n^2$$

$$= 1^2-2^2+3^2-4^2+\dots+(-1)^{n+1}n^2$$

$$= (1-2)(1+2) + (3-4)(3+4) + \dots + (n-1-n)(n+1+n)$$

$$= -(1+2) - (3+4) - \dots - (n+1+n)$$

$$= -(1+2+3+4+\dots+n+1+n)$$

R.H.S.

$$(-1)^{\text{odd}} (1+2+\dots+n)$$

$$\boxed{\text{L.H.S.} = \text{R.H.S.}}$$

proved.)

Q3

$$1 = 1$$

$$3 + 5 = 8$$

$$7 + 9 + 11 = 27$$

$$13 + 15 + 17 + 19 = 64$$

$$21 + 23 + \dots + 29 = 125$$

$$(n^2 - n + 1) + (n^2 - n + 3) + \dots + [n^2 - n + (2n - 1)] = n^3$$

$$\text{L.H.S.} = n(n^2 - n) + (1 + 3 + 5 + \dots + (2n - 1))$$

$$= n^3 - n^2 + (n^2)$$

$$= n^3 - n^2 + n^2$$

$$\therefore \text{L.H.S.} = n^3$$

Chapter-4 Problems of Logic

4.1.3) Six people, named A, B, C, D, E, F are in the dining car of a train. They are one each from NYC, Chicago, Tulsa, St. Louis, Milwaukee and Atlanta. The following facts are ~~are~~ known:

- 1) A and the man from NYC are physicians.
- 2) E and the woman from Chicago are engineers.
- 3) The person from Tulsa and C are engineers.
- 4) B and F are veterans of Gulf War, but the person from Tulsa has never served in military.
- 5) The person from Milwaukee is older than A.
- 6) The person from Atlanta is older than C.
- 7) At St. Louis, B and the man from NYC get off.
- 8) At San Francisco, C and the man from Milwaukee get off.

Match the names of the people with their professions and their cities.

	A	B	C	D	E	F
NYC	X	X	X	#	X	✓
Chicago	X	✓	X	#	X	#
Tulsa	X	X	X	✓	X	X
St Louis	#	#	✓	#	#	#
Milwaukee	X	#	X	#	✓	#
Atlanta	✓	#	X	#	#	#

A = Physician.

Atlanta

E = Teacher

Milwaukee

C = Engineers.

St. Louis.

B = ~~Veterans~~ Teacher

Chicago.

F = ~~Veterans~~ Physician

NYC.

D = Engineer

Tulsa.

We put 'X' in a box that connection is impossible. for eg: 1 guarantees that A is not the person from NYC, so we put 'X' in A column in NYC.

We put '#' in a box that connection is ^{for other} ~~can't be~~ possible. for eg: Since, C is from St. Louis others A, B, D, E, F can't be from St. Louis so we put '#' in those boxes.

We put '✓' in a box that connection is correct for eg: C is '✓' for St. Louis so we put '✓' for C.

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4.2.1

14.1.6

4.4.1

$$\begin{array}{r} \text{L E T S} \\ + \text{W A V E} \\ \hline \text{L A T E R} \end{array}$$

In this problem, the sum of two four digit numbers is 5 digit numbers. So, 1 is carry. Its value is exactly 1.

$$\begin{array}{r} 1 \text{ E T S} \\ + \text{W A V E} \\ \hline 1 \text{ A T E R} \end{array}$$

$W + 1 = A$ and the sum must generate carry. For this W should be 9 or 8.

$$W = 9$$

$$\begin{array}{r} 1 \text{ E T S} \\ + 9 \text{ 0 V E} \\ \hline 1 \text{ 0 T E R} \end{array}$$

Here, $E + 0 = T$, This is only possible only when sum of T and V gives carry.

$$\text{Take } E = 2, T = 3$$

$$\begin{array}{r} \quad \quad \quad 1 \\ \quad \quad \quad \curvearrowright \\ \begin{array}{r} 1 \quad 2 \quad 3 \quad S \\ + 9 \quad 0 \quad V \quad 2 \\ \hline 1 \quad 0 \quad 3 \quad 2 \quad R \end{array} \end{array}$$

Then possible of V is only 8.

We need carry on adding S & 2. So we have 8 & 9 for S which are already taken. So, 2 is not the case for B .

→ Take $B = 3$, then $T = 4$

$$\begin{array}{r} 1 \quad 3 \quad 4 \quad S \\ + 9 \quad 0 \quad V \quad 3 \\ \hline 10 \quad 4 \quad 3 \quad R \end{array}$$

V must be 8 and $S = \{7, 8, 9\}$

If $S = 7$, then $R = 0$, already value of A .

→ Take $B = 4$, $T = 5$,

$$\begin{array}{r} 1 \quad 4 \quad 5 \quad S \\ + 9 \quad 0 \quad V \quad 4 \\ \hline 10 \quad 5 \quad 4 \quad R \end{array}$$

V must be 8 and $S = \{6, 7, 8, 9\}$

If $S = 7$, then $R = 1$, already value of A .

By similar fashion of above, B can't be 4.

→ Take $E=5, T=6$.

$$\begin{array}{r} 1 \quad 5 \quad 6 \quad 5 \\ + 9 \quad 0 \quad V \quad 5 \\ \hline 10 \quad 6 \quad 5 \quad R \end{array}$$

V must be 8 and $S = \{5, 6, 7, 8, 9\}$

If $S=7$, then $R=2$ and from

1. final solution

$$\begin{array}{r} 1 \quad 5 \quad 6 \quad 7 \\ + 9 \quad 0 \quad 8 \quad 5 \\ \hline 10 \quad 6 \quad 5 \quad 2 \end{array}$$

$$\text{NUDE} + \text{NOT} + \text{RUDE} + \text{NOR} = \text{CRUDE.}$$

$$\Rightarrow \text{NUDE} + \text{NOT} + \text{RUDE} + \text{NOR} = C \times 10000 + \text{RUDE.}$$

$$\Rightarrow \text{NUDE} + \text{NOT} + \text{NOR} = C \times 10000$$

The sum of terms of L.H.S. is almost 12000.

So,

C must be 1.

$$\text{NUDE} + \text{NOT} + \text{NOR} = 10,000$$

If 'N' is 9 then sum of L.H.S. is greater than 10000,

If 'N' is less than 8 then sum of L.H.S. is less than 10000,

So, 'N' must be 8.

$$8\text{UDE} + 8\text{OT} + 8\text{OR} = 10000.$$

$$8350 + 826 + 824 = 10000.$$

$$\therefore N=8, U=3, D=5, E=0, O=2, T=6, R=4, C=1.$$

So,

$$8350 + 826 + 4350 + 824 = 14350$$

$$(16) (ATOM)^{172} = A + TO + M$$

$$ATOM = (A + TO + M)^2$$

$$(A + TO + M)^2 = 32^2 \neq (1 + 02 + 4)^2$$

$$\cancel{A + TO + M = 32} =$$

$$33^2 = 1089 \neq (1 + 08 + 9)^2$$

$$(34)^2 = 1156 \neq (1 + 15 + 6)^2$$

$$(35)^2 = 1225 \neq (1 + 22 + 5)^2$$

$$(36)^2 = 1296 \neq (1 + 29 + 6)^2$$

$$\therefore A = 1, T = 2, O = 9, M = 6$$

* Recreational Maths: Magic Square (MATHS)

3x3 Magic Square (MATHS) = MATHS

$$S(1+2+3+4+5+6+7+8+9) = 28 \cdot 1-9 = 107-11$$

$$= 28 \cdot 1-9 = 107-11$$

$$(1+2+3+4+5+6+7+8+9) = 28 \cdot 1-9 = 107-11$$

let S be the sum of the integers of each row.

$$(1+2+3+4+5+6+7+8+9) = 28 \cdot 1-9 = 107-11$$

$$\text{So, } (1+2+3+4+5+6+7+8+9) = 28 \cdot 1-9 = 107-11$$

$$3S = 1+2+3+4+5+6+7+8+9 = 45$$

$$3S = 45$$

$$\therefore S = 15$$

→ Fill 1 at the middle of first row.

→ Fill all 3-periodic numbers by upleft rule.

	1	
		7
4		

→ Remaining all numbers are taken by upright rule.

8	1	6
3	5	7
4	9	2

5X5 magic square.

Let S be sum of the integers of each row.

So,

$$5S = 1 + 2 + \dots + 25$$

$$S = \frac{25 \times 26}{2 \times 5}$$

$$\therefore S = 65$$

→ Fill 1 at the middle of 1st row.

→ Fill all 5-periodic numbers by upright rule ←

		1		
			21	
				16
11				
	6			

1 6 11
16 21

→ Remaining by upright rule.

→

10	18	1	14	22
17	5	13	21	9
4	12	25	8	16
11	24	7	20	3
23	6	19	2	15

for 4×4 matrix.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

for 6x6 matrix.

$$6S = \frac{36 \times 37}{2}$$

$$6S = 666$$

$$S = 111$$

Box 1

Box 2

8	1	6
3	5	7
4	9	2

17	10	15
12	14	16
13	18	11

Box 3

Box 4

26	19	24
21	23	25
22	27	20

36	28	33
30	32	34
31	36	29

Box 1

Box 2

~~Box 3~~

8	1	6	17	10	15
3	5	7	12	14	16
4	9	2	13	18	11
26	19	24	36	28	33
21	23	25	30	32	34
22	27	20	31	36	29

Box 1

Box 3

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8	1	6	26	19	24
3	5	7	21	23	25
4	9	2	22	27	20
35	28	33	12	10	15
30	32	34	12	14	16
31	36	29	13	18	11

Box 2

Box 4

Now,

$$a_{11} \leftrightarrow a_{42}, a_{31} \leftrightarrow a_{61}, a_{02} \leftrightarrow a_{52}$$

35	1	6	26	19	24
3	32	7	21	23	25
31	9	2	22	27	20
8	28	33	12	10	15
30	5	34	12	14	16
4	36	29	13	18	11

For 8×8 matrix.

let, S be the sum of each row or column of the magic square.

$$8S = 1 + 2 + \dots + 64$$

$$8S = \frac{64 \times 65}{2}$$

$$\therefore 8S = 2080$$

$$\therefore S = 260$$

	Box 1				Box 2			
	1	29	33	61	5	25	37	57
	2	30	34	62	6	26	38	58
	3	31	35	63	7	27	39	59
	4	32	36	64	8	28	40	60
	9	21	41	53	13	17	45	49
	10	22	42	54	14	18	46	50
	11	23	43	55	15	19	47	51
	12	24	44	56	16	20	48	52

Box 3

Box 4

Harek box ko 1st Column ma serially plot
gaddai jane. (Box 1 to Box 4 jane)

Next ma harek box ko 2nd Column ma serially
plot gaddai jane (Box 4 to Box 1 jane)

Now, harek box ko diagonals interchange
garney.

Box 1:

$$a_{11} \leftrightarrow a_{44}, a_{22} \leftrightarrow a_{33}$$

$$a_{41} \leftrightarrow a_{34}, a_{32} \leftrightarrow a_{23}$$

Box 2:

$$a_{15} \leftrightarrow a_{48}, a_{45} \leftrightarrow a_{18}$$

$$a_{26} \leftrightarrow a_{37}, a_{36} \leftrightarrow a_{27}$$

Box 3:

$$a_{51} \leftrightarrow a_{84}, a_{62} \leftrightarrow a_{73}$$

$$a_{81} \leftrightarrow a_{54}, a_{63} \leftrightarrow a_{72}$$

Box 4:

$$a_{55} \leftrightarrow a_{88}, a_{65} \leftrightarrow a_{78}$$

$$a_{86} \leftrightarrow a_{57}, a_{66} \leftrightarrow a_{77}$$

64	29	33	4	60	25	37	8
2	35	31	62	6	39	27	58
3	34	30	63	7	38	26	59
61	32	36	1	50	28	40	5
56	21	41	12	52	17	45	16
10	43	23	54	14	47	19	50
11	42	22	55	15	46	18	57
53	24	44	9	49	20	48	13

Welghing problems.

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5.2.1) 9 pearls problems, all look same, 1 has different weight (odd pearl). Find odd pearl in just 3 weighings.

→ let us divide 9 pearls into 3 equal groups, each has 3 pearls.

Let the groups be $G_1 = \{a_1, b_1, c_1\}$, $G_2 = \{a_2, b_2, c_2\}$ and $G_3 = \{a_3, b_3, c_3\}$

(1st Weighing) Weigh G_1 against G_2 . We get three cases,

Case I: $G_1 = G_2$

Case II: $G_1 > G_2$

Case III: $G_1 < G_2$

Let, $G_1 = G_2$, Then all 6 pearls of G_1 and G_2 are normal pearls, and odd pearl is in G_3 .

(2nd Weighing) : Weigh a_3 against b_3 .

→ If $a_3 = b_3$, then odd pearl is c_3 .

(3rd Weighing) : Weigh c_3 with / (against) a_1 .

- If $c_3 < a_1$, then c_3 is light.
- If $c_3 > a_1$, then c_3 is heavy.

(2nd)

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→ If $a_3 < b_3$ then any of these two is odd pearl, but not c_3 .

(3rd)

• a_3 against a_1 .

- If $a_3 = a_1$, then b_3 is odd pearl and it is heavier.
- If $a_3 < a_1$, then a_3 is odd pearl and it is lighter.

Similarly • $\Rightarrow a_3 > b_3$.

Case II: $G_1 < G_2$.

• Then odd pearl is in either G_1 or in G_2 .

(2nd Weighing)

• Weight G_1 against G_3 .

→ if $G_1 = G_3$, then odd pearl is in G_2 , and the odd pearl is heavier.

(3rd weighing)

a_2 vs b_2 .

- if $a_2 = b_2$ then odd pearl is c_2 .
- if $a_2 < b_2$ then odd pearl is b_2 .
- if $a_2 > b_2$ then odd pearl is a_2 .

(2nd Weighing)

if $\rightarrow (G_1 < G_3)$, then odd pearl is in G_1 and it is lighter.

(3rd Weighing):

Weight a_1 against b_1 .

- $a_1 = b_1$ then c_1 is odd pearl.
- $a_1 > b_1$ then b_1 is odd pearl.
- $a_1 < b_1$ then a_1 is odd pearl.

Similarly, we can find odd pearl from case III.

5.25) You have 80 pearls. One is lighter than all the others. Find the odd pearl in just four weighings on a balance scale.

Solⁿ.

Let us make 3 groups of pearls containing, 27, 27, 26, let the groups are G_1, G_2, G_3 .

1st Weighing

Weigh G_1 against G_2 . We get three cases.

Case I :- $G_1 = G_2$, Case II, $G_1 < G_2$, Case III; $G_1 > G_2$

Case I :-

$G_1 = G_2$.

Odd pearl is in G_3 .

Again, Divide the pearls of G_3 into 3 groups having, 9, 9, 8

2nd Weighing :- Weigh two groups which has 9 pearls.

Sub Case I :- If they are balanced, odd pearls is in the group of 8.

Again divide the pearls of group 8 into three subgroups having 3, 3, 2.

3rd Weighing: Weigh two groups of 3 pearls, if they are balanced then odd pearl is in the group of 2.

4th Weighing: Weigh 2 pearls, the lighter one is odd pearl.

Subcase II: If the groups of 9 and 9 are not balanced, then odd pearl is either of this two groups.

3rd weighing: Divide 9 pearls into three equal groups containing 3 pearls. Weigh any two groups.

- If they balance, then odd pearl is in third group.
- If they do not balance, then odd pearl is in either of this two groups.

4th Weighing: (not balanced) weigh any two of third group if the balance, next one will be odd pearl. otherwise the lighter one is odd pearl.

Chapter - 6
Algebra and Analysis

6.1.3) If n is positive integer, show that $n^3 - n$ is divisible by 3.
Solⁿ.

Here,

$$\begin{aligned} n^3 - n \\ n(n^2 - 1) \\ n(n+1)(n-1) \end{aligned}$$

These three factor are three consecutive integer and any of which is divisible by 3.

6.1.2) $n^5 - n$ is divisible by 5.

Solⁿ.

$$\begin{aligned} n^5 - n \\ n(n^4 - 1) \\ n(n^2 + 1)(n^2 - 1) \\ n(n^2 + 1)(n-1)(n+1) \end{aligned}$$

5 divides those number whose last digit is either 0 or 5.

- If last digit is 0 or 5, then 5 divides n .
- If last digit is 1 or 6, then 5 divides $(n-1)$.
- If last digit is 4 or 9, then 5 divides $n+1$.
- If last digit is 2, 3, 7, 8, then 5 divides $n^2 + 1$.
- divides $n^5 - n$ for all integers.

6.3.4.

$$\binom{k}{m} + \binom{k}{m+1} = \binom{k+1}{m+1}$$

$$\text{L.H.S.} = \frac{k!}{(k-m)!m!} + \frac{k!}{(k-(m+1))!(m+1)!}$$

$$\frac{k!}{(k-m)!m!} + \frac{k!}{(k-(m+1))!(m+1)!}$$

$$\frac{k! (m+1)}{(k-m)!m!(m+1)!} + \frac{k! (k-m)}{((k-m-1))!(m+1)!(k-m)}$$

$$\frac{k! (m+1)}{(k-m)!(m+1)!} + \frac{k! (k-m)}{(k-m)!(m+1)!}$$

$$\frac{k! (m+1) + k! (k-m)}{(k-m)!(m+1)!}$$

$$\frac{k! (m+1 + k-m)}{(k-m)!(m+1)!}$$

$$\frac{k! (k+1)}{((k+1)-(m+1))!(m+1)!}$$

$$\frac{(k+1)!}{((k+1)-(m+1))!(m+1)!} = \binom{k+1}{m+1}$$

Binomial Expansion.

$$(a+b)^1 = a+b$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a+b)^n = [a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + b^n]$$

$$n^{\text{th}} \text{ term} = {}^nC_r a^{n-r} b^r$$

6.1.6) $\alpha = (1 + 0.000001)^{1000000}$ or 2 greater?

$$= 1^{1000000} + \binom{1000000}{1} 1^{999999} (0.000001) + \binom{1000000}{2} 1^{999998} (0.000001)^2 + \dots + (0.000001)^{1000000}$$

$$= 1 + 1000000 * 0.000001 + \text{positive value.}$$

$$= 1 + 1 + \text{positive value.}$$

$$\alpha = 2 + \text{positive value.}$$

$$\therefore \alpha > 2.$$

Which is greater?

1000^{1000} or 1001^{999}

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Solⁿ.

$$(1000 + 1)^{999} = (1000 + 1)^{999}$$

$$= 1000^{999} + \binom{999}{1} 1000^{998} \cdot 1 + \binom{999}{2} 1000^{997} \cdot 1^2 + \dots + 1^{999}$$

$$< 1000^{999} + 1000 \times 1000^{998} + 1000^2 \cdot 1000^{997} + \dots + 1000^{999}$$

$$= 1000^{999} + 1000^{999} + 1000^{999} + \dots + 1000^{999}$$

$$= 1000 \cdot 1000^{999}$$

$$= 1000^{1000}$$

$$\therefore (1001)^{999} < 1000^{1000}$$

6.1.8) Assume k is an integer.
Calculate,

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)}$$

$$= \left(\frac{1}{2} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

$$= 1 - \frac{1}{k+1} = \frac{k}{k+1}$$

6.1.8.) $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \dots + \frac{1}{(k-2)(k-1)k}$

$$= \sum_{n=3}^k \frac{1}{(n-2)(n-1)n}$$

$$= \frac{1}{2} \sum_{n=3}^k \frac{n - (n-2)}{(n-1)(n-2)n}$$

$$= \frac{1}{2} \sum_{n=3}^k \left[\frac{1}{(n-2)(n-1)} - \frac{1}{(n-1)n} \right]$$

$$= \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} \right] + \left(\frac{1}{2 \cdot 3} - \frac{1}{3 \cdot 4} \right) + \dots + \left(\frac{1}{(k-2)(k-1)} - \frac{1}{(k-1)k} \right)$$

$$= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{(k-1)k} \right]$$

$$= \frac{1}{4} \left[\frac{k^2 - k - 2}{k(k-1)} \right] = \frac{1}{4} \frac{(k-2)(k+1)}{k(k-1)}$$

6.1.30) Calculate

$$1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1)$$

Solⁿ,

We know,

$$1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1)$$

$$\sum_{n=1}^k n(n+1)$$

$$\sum_{n=1}^k (n^2 + n)$$

$$\sum_{n=1}^k n^2 + \sum_{n=1}^k n$$

$$= \frac{k(k+1)(2k+1)}{6} + \frac{k(k+1)}{2}$$

$$= \frac{k(k+1)}{2} \left[\frac{2k+1}{3} + 1 \right]$$

$$= \frac{k(k+1)}{2} \times \frac{2(k+2)}{3} = \frac{k(k+1)(k+2)}{3}$$

$$6.1.1) 1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + k(k+1)(k+2)$$

We have,

$$\sum_{n=1}^k n(n+1)(n+2)$$

$$\sum_{n=1}^k (n^2 + n)(n+2) \Rightarrow \sum_{n=1}^k n^3 + 2n^2 + n^2 + 2n$$

$$= \sum_{n=1}^k n^3 + 3 \sum_{n=1}^k n^2 + 2 \sum_{n=1}^k n$$

$$= \frac{n^2(n+1)^2}{4} + \frac{3 \cdot n(n+1)(2n+1)}{6} + \frac{2n(n+1)}{2}$$

$$= \frac{n^2(n^2+2n+1)}{4} + n^2$$

$$= \frac{n^2(n+1)}{2} \left[\frac{n(n+1)}{2} + (2n+1) + 2 \right]$$

$$= \frac{n(n+1)}{4} [n(n+1) + 2(2n+1) + 4]$$

$$= \frac{n(n+1)}{4} [n^2 + n + 4n + 2 + 4]$$

$$= \frac{n(n+1)}{4} [n^2 + 5n + 6] = \frac{n(n+1)(n+2)(n+3)}{4}$$

6.2.4) If a and b are positive real numbers, show that

$$ab \leq \frac{a^2 + b^2}{2}$$

Solⁿ.

For any two positive real numbers a and b ,

$$(a-b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0$$

$$a^2 + b^2 \geq 2ab$$

$$\frac{a^2 + b^2}{2} \geq ab \quad \text{proved}$$

$$\log_a x = \frac{\ln x}{\ln a}$$

$$a \ln b = \ln b^a$$

$$\ln(a \cdot b) = \ln a + \ln b$$

$$\ln(a \cdot b) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

6.2.3)

Show that,

$$2 < \frac{1}{\log_2 \pi} + \frac{1}{\log_5 \pi}$$

$$2 < \frac{1}{\frac{\ln \pi}{\ln 2}} + \frac{1}{\frac{\ln \pi}{\ln 5}}$$

$$2 < \frac{\ln 2 + \ln 5}{\ln \pi}$$

$$2 < \frac{\ln(2 \times 5)}{\ln \pi}$$

$$(\ln \pi)^2 < \ln(10)$$

$$\pi^2 < 10, \quad \text{which is true.}$$

6.2.4) Show that:

$$2 < \frac{1}{\log_2 \pi} + \frac{1}{\log_{\pi} 2}$$

$$< \frac{\ln 2}{\ln \pi} + \frac{\ln \pi}{\ln 2}$$

$$\frac{(\ln 2)^2 + (\ln \pi)^2}{\ln \pi \cdot \ln 2}$$

$$2 \ln(2 \cdot \pi)$$

$$< \frac{(\ln 2 + \ln \pi)^2}{2 \ln 2 \ln \pi}$$

$$2(\ln \pi \cdot \ln 2) = (\ln 2 - \ln \pi)^2 + 2 \ln 2 \cdot \ln \pi$$

or $0 < (\ln 2 - \ln \pi)^2$ which is always positive

6.2.5.

Show,

$$|\cos x + \sin x| \leq \sqrt{2}$$

We know,

$$|x| = \sqrt{x^2}$$

So,

$$\begin{aligned} |\cos x + \sin x| &= \sqrt{(\cos x + \sin x)^2} \\ &= \sqrt{\cos^2 x + 2\sin x \cdot \cos x + \sin^2 x} \\ &= \sqrt{1 + \sin 2x} \end{aligned}$$

$\sin 2x \in [-1, 1]$, $\sqrt{1 + \sin 2x}$ is maximum when $\sin 2x = 1$, So,

$$|\cos x + \sin x| \leq \sqrt{1+1} = \sqrt{2} \text{ proved,}$$

$$|\cos x - \sin x| \leq \sqrt{2}$$

$\frac{\sin}{\csc}$	A
$\frac{\tan}{\cot}$	$\frac{\cos}{\sec}$

6.2.2) Which is greater?

$\sin(\cos x)$ or $\cos(\sin x)$?

We know,

$$\sin \theta = -\cos \left(\frac{\pi}{2} + \theta \right)$$

$$\therefore \sin(\cos x) = -\cos \left(\frac{\pi}{2} + \cos x \right)$$

Now,

$$\cos(\sin x) - \sin(\cos x) = \cos(\sin x) + \cos \left(\frac{\pi}{2} + \cos x \right)$$

$$= 2 \cos \left(\frac{\sin x + \frac{\pi}{2} + \cos x}{2} \right) \cdot \cos \left(\frac{\frac{\pi}{2} + (\cos x - \sin x)}{2} \right)$$

$$\leq 2 \cos \left(\frac{\sqrt{2} + \pi/2}{2} \right) \cdot \cos \left(\frac{\sqrt{2} + \pi/2}{2} \right)$$

$$= 2 \cos^2 \left(\frac{\sqrt{2} + \pi/2}{2} \right)$$

= positive quantity.

$$\therefore \cos(\sin x) > \sin(\cos x).$$

6.3.5

Show,

$$\frac{\cos \alpha}{2} \cdot \frac{\cos \alpha}{4} \cdot \frac{\cos \alpha}{8} = \frac{\sin \alpha}{8 \sin \left(\frac{\alpha}{8} \right)}$$

$$\text{or, } \frac{\cos \alpha}{2} \cdot \frac{\cos \alpha}{4} \cdot \frac{\cos \alpha}{8} \cdot 8 \sin \left(\frac{\alpha}{8} \right) = \sin \alpha$$

$$\text{or, } \frac{2 \cos \alpha}{2} \cdot \frac{2 \cos \alpha}{4} \cdot \frac{2 \cos \alpha}{8} \cdot \frac{\sin \alpha}{8} = \sin \alpha$$

$$\text{or, } \frac{2 \cos \alpha}{2} \cdot \frac{2 \cos \alpha}{4} \cdot \frac{\sin \alpha}{4} = \sin \alpha$$

$$\text{or, } \frac{2 \cos \alpha}{2} \cdot \frac{\sin \alpha}{2} = \sin \alpha$$

$$\therefore \sin \alpha = \sin \alpha$$

✓

Exercise 6.

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1) If $0 \leq a, b, c, d \leq 1$, Show,

Soln.

$$\begin{aligned}(1-a)(1-b)(1-c)(1-d) &= \cancel{1-a-c-d-b} \\ &\quad 1-c-d+cd-a+ac \\ &\quad +ad-aed-b+bc \\ &\quad +bd-bcd+ab \\ &\quad -abc-abd+abcd \\ &= (1-a-b-c-d) + ab(1-c) + ac(1-d) \\ &\quad + ad(1-b) + \cancel{1} + bd(1-a) \\ &\quad + ad+cd+abcd.\end{aligned}$$

Here,

$$(1-c)(1-d), (1-b), (1-a) \geq 0.$$

So, the all terms of R.H.S except $1-a-b-c-d$ are non-negative.

$$\therefore (1-a)(1-b)(1-c)(1-d) \geq 1-a-b-c-d.$$

2) If a, b, c, d are +ve real numbers, show,

$$\frac{(a^2+1)(b^2+1)(c^2+1)(d^2+1)}{abcd} \geq 16.$$

Solⁿ

for any positive real number, a ,

$$(a-1)^2 \geq 0$$

$$\therefore a^2 - 2a + 1 \geq 0$$

$$\therefore a^2 + 1 \geq 2a$$

$$\therefore \frac{a^2 + 1}{a} \geq 2$$

Similarly, for b, c, d ,

$$\frac{b^2+1}{b} \geq 2, \quad \frac{c^2+1}{c} \geq 2, \quad \frac{d^2+1}{d} \geq 2$$

So,

$$\left(\frac{a^2+1}{a}\right) \left(\frac{b^2+1}{b}\right) \left(\frac{c^2+1}{c}\right) \left(\frac{d^2+1}{d}\right) \geq 2 \cdot 2 \cdot 2 \cdot 2$$

$$\therefore \frac{(a^2+1)(b^2+1)(c^2+1)(d^2+1)}{abcd} \geq 16.$$

fix

(5) Show that $11^{10} - 1$ is divisible by 100.

Sol.

100 divides those numbers whose last two digits are zero.

$$\begin{aligned} 11^{10} - 1 &= (10+1)^{10} - 1 \\ &= \left[10^{10} + \binom{10}{1} 10^9 \cdot 1 + \binom{10}{2} 10^8 \cdot 1^2 + \dots + \binom{10}{9} 10^1 \cdot 1^9 + \binom{10}{10} 10^0 \cdot 1^{10} \right] - 1 \end{aligned}$$

$$\begin{aligned} &= \left[10^{10} + 10^9 + \binom{10}{2} 10^8 + \dots + \binom{10}{8} \cdot 10^2 + 10 \cdot 10 \right] + 1^{10} - 1 \end{aligned}$$

$$\therefore 11^{10} - 1 =$$

Each term of has two ^{digits} ~~terms~~ as zero so, every number is divisible by 100. So, this number is divisible by 100.